



# **SJB INSTITUTE OF TECHNOLOGY**

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## **DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING**

### **DIGITAL SIGNAL PROCESSING (IPCC21EC42)**

#### **MODULE-2**

### **ADDITIONAL DFT PROPERTIES, LINEAR FILTERING METHODS BASED ON THE DFT**

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## FILTERING OF LONG DATA SEQUENCE:-

In practical applications involving linear filtering of signals, the input sequence  $x(n)$  is often a very long sequence.

Since linear filtering performed via the DFT involves operations on a block of data, which must be limited in size due to limited memory of digital computer.

Hence a long sequence must be segmented to fixed-size blocks before processing. Each block is processed via the DFT and IDFT to produce a block of output data. The output blocks are fitted together to form an overall output sequence which is identical to the sequence obtained if the long block has been processed via time-domain convolution.

Two methods

① Overlap-save method

② Overlap-add method

Both methods assume impulse resp  $h(n)$  of length  $m$ . If data segmented into blocks of  $L$  points where  $L \gg m$ .

## Overlap-save method:-

- Size of input data blocks is  $N = L + m - 1$  and DFT's and IDFTs are of length  $N$ .
- Out of the  $N$  elements first  $m - 1$  points are from the previous data block followed by  $L$  new data points
- ~~$N$ -point DFT is computed for each block, with  $h(m)$  (impulse response of FIR filter) increased in length to  $N$  by appending  $L - 1$  zeros ( $N = L + m - 1$ ) and taking  $N$ -pt DFT.~~
- $N$ -point DFT is computed for each block.
- Impulse response of FIR filter  $h(m)$  is increased in length to  $N$  by appending  $L - 1$  zeros. The  $N$ -pt DFT is computed once and stored.
- The multiplication of two  $N$ -pt DFT's  $\{H(k)\}$  and  $\{x_m(k)\}$  for the  $m$ th block of data yields

$$\hat{Y}_m(k) = H(k) x_m(k) \quad k = 0, 1, \dots, N-1$$

→ (P)

Then  $N$ -pt IDFT yields the result

$$\hat{y}_m(m) = \left\{ \hat{y}_m(0) \hat{y}_m(1) \dots \hat{y}_m(m-1) \hat{y}_m(m) \dots \hat{y}_m(N-1) \right\} \quad - (2)$$

Since data record is of length  $N$  the first  $m-1$  points of  $y_m(m)$  are corrupted by aliasing and must be discarded.

The last  $L$  points of  $y_m(m)$  are exactly the same as the result from linear convolution

Hence,

$$\hat{y}_m(m) = y_m(m), \quad m = m, m+1, \dots, N-1 \quad - (3)$$

- To avoid loss of data due to aliasing the last  $m-1$  points of each data record are saved and these points become the first  $m-1$  data points of the subsequent record. (3)

- To begin processing, the first  $m-1$  points of the first record are set to zero. Thus blocks of data sequence are

$$x_1(m) = \left\{ \underbrace{0, 0, \dots, 0}_{m-1 \text{ points}}, x(0), x(1) \dots x(L-1) \right\} \quad - (4)$$

$$x_2(m) = \left\{ \underbrace{x(L-m+1) \dots x(L-1)}_{m-1 \text{ data points from } x_1(m)}, \underbrace{x(L) \dots x(2L-1)}_{L - \text{new data points}} \right\} \quad - (5)$$

$$x_3(m) = \underbrace{x(2L-m+1), \dots, x(2L-1)}_{m-1 \text{ data points from } x_2(m)}, \underbrace{x(2L), \dots, x(3L-1)}_{L \text{ new data points.}}$$

$\rightarrow$  so on.

The resulting data sequence from the IDFT is  $y_3(m)$  where first  $m-1$  points are discarded due to aliasing and remaining  $L$  points constitute the desired result from linear convolution.

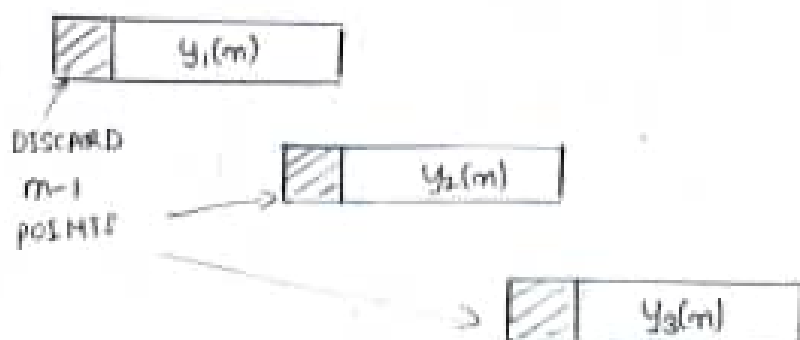
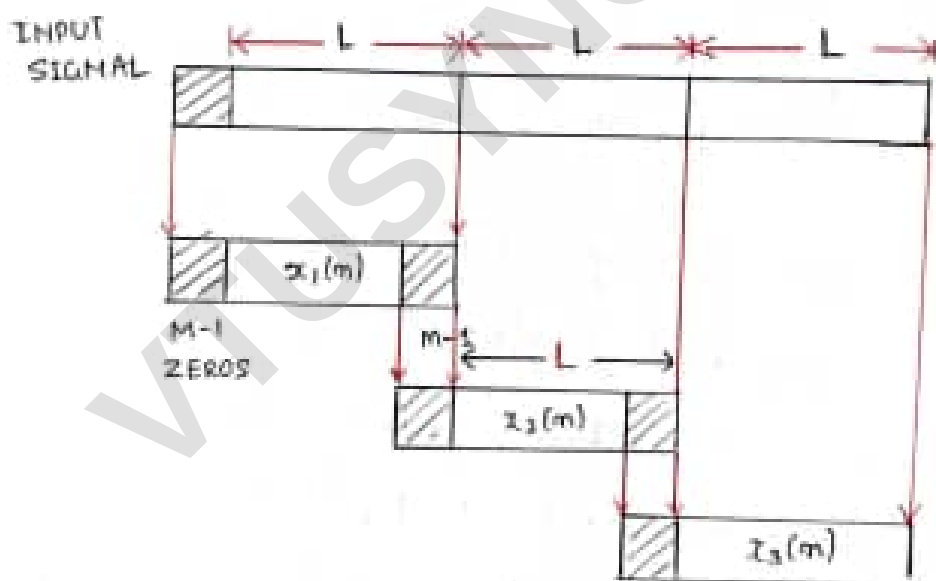


Fig Linear FIR filtering by the overlap-save method



(P) Find the o/p  $y(n)$  of a filter whose impulse response is  $h(n) = \{1, 2\}$  and the input signal to the filter is  $x(n) = \{1, 2, -1, 2, 3, -2, -3, -1, 1, 1, 2, -1\}$  using overlap-save method. (JUN 2010 / 8M)

Soln:- Given  $h(n) = \{1, 2\}$

$x(n) = \{1, 2, -1, 2, 3, -2, -3, -1, 1, 1, 2, -1\}$

W.K.T  $N = L + m - 1$

$L$  = Block length of i/p data sequence

$m$  = Length of impulse response  $h(n)$

$N$  = Size of DFT + IDFT or convolution length.

Given  $m = 2$

$$\therefore N = 2^m = 2^2 = 4$$

$$N = L + m - 1$$

$$4 = L + 2 - 1$$

$\Rightarrow$

$$\boxed{L = 3}$$

Given data sequence is segmented into blocks of length  $L = 3$ .

$$x_1(n) = \{1, 2, -1\}$$

$$x_2(n) = \{2, 3, -2\}$$

$$x_3(n) = \{-3, -1, 1\}$$

$$x_4(n) = \{1, 2, -1\}$$

$$x_5(n)$$

(11)

Since we want to perform 4-point circular convolution, padding  $h(m)$  with zeros

$$h(m) = \{1, 2, 0, 0\}$$

In ~~over~~ overlap-save method, first subsequence will have first  $(m-1)$  zeros and next  $L$  i/p samples.

Modified subsequences are

$$x_1(m) = \{0, 1, 2, -1\}$$

$$x_2(m) = \{-1, 2, 3, -2\}$$

$$x_3(m) = \{-2, -3, -1, 1\}$$

$$x_4(m) = \{1, 1, 2, -1\}$$

$$x_5(m) = \{-1, 0, 0, 0\}$$

To find o/p of each subsequence

$$y_1(m) = x_1(m) \oplus h(m) = \{-2, 1, 4, 3\}$$

$$y_2(m) = x_2(m) \oplus h(m) = \{-5, 0, 7, 4\}$$

$$y_3(m) = x_3(m) \oplus h(m) = \{0, -7, -7, -1\}$$

$$y_4(m) = x_4(m) \oplus h(m) = \{-1, 3, 4, 3\}$$

$$y_5(m) = x_5(m) \oplus h(m) = \{-1, -2, 0, 0\}$$

$$y_1(m) = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ -1 \end{bmatrix} = [-2, 1, 4, 3]$$

$$y_2(m) = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 3 \\ -2 \end{bmatrix} = [-5, 0, 7, 4]$$

$$y_3(m) = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ -3 \\ -1 \\ 1 \end{bmatrix} = [0, -7, -7, -1]$$

$$y_4(m) = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \\ -1 \end{bmatrix} = [-1, 3, 4, 3]$$

$$y_5(m) = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = [-1, -2, 0, 0]$$

(13)

Combining all outputs we get final o/p



$$y_1(m) = \boxed{-2}, 1, 4, 3$$

$$y_2(m) = \begin{matrix} \text{DISCARD} \\ m-2 \end{matrix} \rightarrow \boxed{-5}, 0, 7, 4$$

$$y_3(m) = \boxed{0}, -7, -7, -1$$

$$y_4(m) = \boxed{-1}, 3, 4, 3$$

$$y_5(m) = \boxed{-1}, 2, 0, 0$$

$$\therefore y(m) = \{ 1, 4, 3, 0, 7, 4, -7, -7, -1, 3, 4, 3, 2, 0, 0 \}$$

is the complete o/p.

- ⑥ The unit impulse response of a certain FIR digital filter is  $h(m) = \frac{1}{2} \{ 1, 2, 1 \}$ . Determine its step response using overlap save method. Sketch the step response. Also verify from the response whether the filter is stable & causal.

Soln:- Given  $h(m) = \frac{1}{2} \{ 1, 2, 1 \}$

(14)

Step response means response of an LTI system to step inp ie  $x(m) = u(m)$

$$u(m) = \{ 1, 1, 1, 1, \dots \}$$

$$m = 3, \quad N = 2^m = 2^3 = 8$$

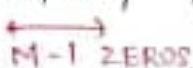
$$N = L + m - 1 \Rightarrow 8 = L + 3 - 1$$

Studied smart, not hard — thanks to VTUSync.in

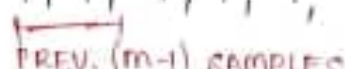
$$\boxed{L = 6}$$

$$\therefore h(m) = \{1, 2, 1, 0, 0, 0, 0, 0\}$$

$$x_1(m) = \{0, 0, 1, 1, 1, 1, 1, 1\}$$



$$x_2(m) = \{1, 1, 1, 1, 1, 1, 1, 1\}$$



$$x_3(m) = \{1, 1, 1, 1, 1, 1, 1, 1\}$$

$\vdots$   
 $\vdots$   
 $\vdots$

$$y_1(m) = x_1(m) \textcircled{8} h(m)$$

$$= \{3, 1, 1, 3, 4, 4, 4, 4\}$$

$$y_2(m) = x_2(m) \textcircled{8} h(m)$$

$$= \{4, 4, 4, 4, 4, 4, 4, 4\}$$

$$y_3(m) = x_3(m) \textcircled{8} h(m)$$

$$= \{4, 4, 4, 4, 4, 4, 4, 4\}$$

Hence o/p sequence is obtained from the subsequence as: — (15)

$$\boxed{3, 1}, 1, 3, 4, 4, 4, 4$$

NEGLECT

$$\boxed{4, 4}, 4, 4, 4, \dots$$

$y(m) = \{1, 3, 4, 4, 4, 4, 4, \dots\}$

- (P) A long sequence  $x(n)$  is filtered through a filter with impulse response  $h(n)$  to yield the o/p  $y(n)$  of
- (1)  $h(n) = \{1, -1\}$  &  $x(n) = \{1, 1, 1, 1, 1, 3, 1, 1, 4, 2, 1, 1, 3, 1\}$ . Compute  $y(n)$  using overlap-save techniques. [use only 5 point circular convolution].

Soln:- Given  $M=2$   
 $N=5$

$$N = L + M - 1$$

$$5 = L + 2 - 1$$

$$\text{or } L = 4$$

Input sequence is divided into subsequences of length  $L=4$ .

$$\text{i.e. } x_1(n) = \{1, 1, 1, 1\}$$

$$x_2(n) = \{1, 3, 1, 1\}$$

$$x_3(n) = \{4, 2, 1, 1\}$$

$$x_4(n) = \{3, 1, 0, 0\}$$

Since we want to perform 5-point circular convolution & using overlap save technique

$$h(n) = \{1, -1, 0, 0, 0\}$$

$$x_1(m) = \{0, 1, 1, 1, 1\}$$

$$x_2(m) = \{1, 1, 3, 1, 1\}$$

$$x_3(m) = \{1, 4, 2, 1, 1\}$$

$$x_4(m) = \{1, 3, 1, 0, 0\}$$

$$y_1(m) = x_1(m) \otimes h(m) = \{-1, 1, 0, 0, 0\}$$

$$y_2(m) = x_2(m) \otimes h(m) = \{0, 0, 2, -2, 0\}$$

$$y_3(m) = x_3(m) \otimes h(m) = \{0, 3, -2, -1, 0\}$$

$$y_4(m) = x_4(m) \otimes h(m) = \{1, 2, -2, -1, 0\}$$

Discarding first  $(M-1)$  samples & arranging.

$$y(m) = \{1, 0, 0, 0, 0, 2, -2, 0, 3, -2, -1, 0, 2, -2, -1, 0\}$$

(14)

### Overlap-add Method: -

In this method the size of the input data block is  $L$  points and the size of DFTs and IDFTs is  $N = L + m - 1$ .

To each data block we append  $m-1$  zeros and compute the  $N$ -point DFT.

Data blocks may be represented as,

$$x_1(m) = \{ x(0), x(1), \dots, x(L-1), \underbrace{0, 0, 0, \dots, 0}_{m-1 \text{ ZEROS}} \}$$

$$x_2(m) = \{ x(L), x(L+1), \dots, x(2L-1), \underbrace{0, 0, \dots, 0}_{m-1 \text{ ZEROS}} \}$$

$$x_3(m) = \{ x(2L), x(2L+1), \dots, x(3L-1), \underbrace{0, 0, \dots, 0}_{m-1 \text{ ZEROS}} \}$$

so on.

The two  $N$ -point DFTs are multiplied together to form

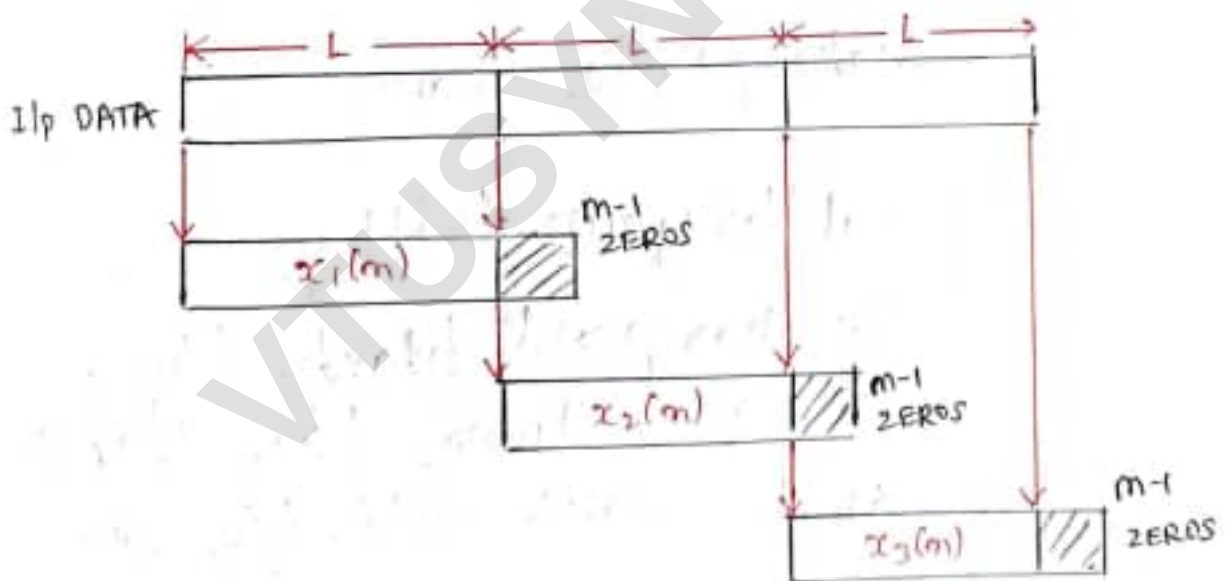
$$Y_m(k) = H(k) X_m(k) \quad k=0, 1, \dots, N-1$$

The IDFT yields data blocks of length  $N$  that are free from aliasing, since the size of the DFTs and IDFTs is  $N = L + m - 1$  and the sequences are increased to  $N$ -points by appending zeros to each block.

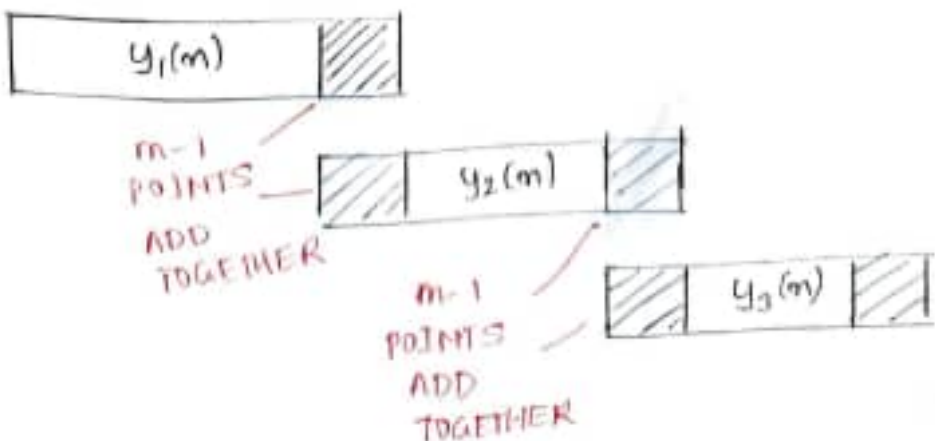
Since each data block is terminated with  $m-1$  zeros, the last  $m-1$  points from each output block must be overlapped and added to the first  $m-1$  points of the succeeding block.

Hence the method is called overlap-add method.

The output sequence,  $y(n) = \{ y_1(0), y_1(1), \dots, y_1(L-1), y_1(L) + y_2(0), y_1(L+1) + y_2(1), \dots, y_1(M-1) + y_2(m-1), y_2(m), \dots \}$



OUTPUT DATA





30) Find the output  $y(n)$  of a filter whose impulse response is  $h(n) = \{1, -2\}$  and input signal  $x(n) = \{3, -2, 4, 1, 5, 7, 2, -9\}$  using overlap-add method. Use only 5-point circular convolution in your approach. (5)

Soln: Given  $x(n) = \{3, -2, 4, 1, 5, 7, 2, -9\}$   
 $h(n) = \{1, -2\}$

WKT  $M=2$ ,  $N=5$

$$N = L + M - 1$$

$$5 = L + 2 - 1$$

$$\boxed{L=4}$$

$\therefore$  we divide the i/p sequences to subsequences of length  $L=4$ .

$$x_1(n) = \{3, -2, 4, 1\}$$

$$x_2(n) = \{5, 7, 2, -9\}$$

using overlap add technique we need to pad  $(M-1)$  zeros at the end of each subsequence.

$$x_1(n) = \{3, -2, 4, 1, \overset{(M-1) \text{ zero.}}{0}\}$$

$$x_2(n) = \{5, 7, 2, -9, 0\}$$

$h(n) = \{1, -2, 0, 0, 0\}$  by padding zeros to make  $h(n)$  of length 'N'.

$$y_1(n) = x_1(n) \otimes h(n)$$

$$y_1(n) = \begin{bmatrix} 3 & 0 & 1 & 4 & -2 \\ -2 & 3 & 0 & 1 & 4 \\ 4 & -2 & 3 & 0 & 1 \\ 1 & 4 & -2 & 3 & 0 \\ 0 & 1 & 4 & -2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -8 \\ 8 \\ -7 \\ -2 \end{bmatrix}$$

$$y_2(n) = x_2(n) \otimes h(n)$$

$$y_2(n) = \begin{bmatrix} 5 & 0 & -9 & 2 & 7 \\ 7 & 5 & 0 & -9 & 2 \\ 2 & 7 & 5 & 0 & -9 \\ -9 & 2 & 7 & 5 & 0 \\ 0 & -9 & 2 & 7 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \\ -12 \\ -13 \\ 18 \end{bmatrix}$$

Adding last  $(M-1)$  samples of  $y_1(n)$  with 1<sup>st</sup>  $(M-1)$  samples of  $y_2(n)$  we get,

$$y_1(n) = \begin{matrix} 3 & -8 & 8 & -7 & -2 \\ & & & & + \end{matrix}$$

$$y_2(n) = \begin{matrix} & & & & 5 & -3 & -12 & -13 & 18 \end{matrix}$$

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$$\begin{matrix} 3 & -8 & 8 & -7 & 3 & -3 & -12 & -13 & 18 \end{matrix}$$

$$\therefore y(n) = \{3, -8, 8, -7, 3, -3, -12, -13, 18\}$$

## EFFICIENT COMPUTATION OF DFT:

### FAST FOURIER TRANSFORM ALGORITHM (FFT)

FROM DFT and IDFT equations

$$X(k) = \sum_{n=0}^{N-1} x(n) (W_N)^{kn} \quad 0 \leq k \leq N-1$$

$$\& x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) (W_N)^{-kn} \quad 0 \leq n \leq N-1$$

From the above two equations we observe that Direct computation of  $X(k)$  involves  $N^2$  complex multiplications &  $4N^2$  Real multiplications &  $N(N-1)$  complex additions.

If  $x(n)$  is complex, we require

$4N^2$  Real multiplications

$4N(N-1)$  Real Additions.

### Properties of Twiddle Factor [Phase Factor]

#### 1) SYMMETRY PROPERTY

$$W_N^{k + \frac{N}{2}} = -W_N^k$$

Proof:-  $W_N^{k + \frac{N}{2}} = W_N^k \cdot W_N^{N/2}$

$$= W_N^k \left( e^{-j \frac{2\pi}{N}} \right)^{N/2}$$

$$= W_N^k [\cos \pi - j \sin \pi]$$

$$= \underline{\underline{-W_N^k}} \quad \text{HENCE THE PROOF.}$$

## 2) PERIODICITY PROPERTY

$$W_N^{k+N} = W_N^k$$

Proof:-

$$W_N^{k+N} = W_N^k \cdot W_N^N$$

$$= W_N^k \left( e^{-j \frac{2\pi}{N}} \right)^N$$

$$= \underline{\underline{W_N^k \cdot 1}}$$

HENCE THE PROOF

PROPERTY PROP:  $W_N^2 = W_{N/2}$

## TYPES OF FFT ALGORITHMS

### 1) Radix-2 FFT algorithm

Total length of DFT is represented as  $2^v$ , where  $v$  is an integer.

$$\text{ie, } N = 2^1, 2^2, 2^3, \dots$$

$$= 2, 4, 8, \dots$$

## 2) Radix-4 FFT Algorithm

Total length  $N = 4^V$

$$\text{ie, } N = 4^1, 4^2, 4^3, \dots \\ = 4, 16, 64, \dots$$

## 3) Radix-8 FFT

Total length  $N = 8^V$

$$\text{ie, } N = 8^1, 8^2, \dots \\ N = 8, 64, \dots$$

## 4) Split-Radix FFT

It is a combination of Radix-2  
and Radix-4.



## Direct Computation of DFT

For a complex valued sequence  $x(n)$  of  $N$  points, DFT may be expressed as.

$$X_R(k) = \sum_{n=0}^{N-1} \left[ x_R(n) \cos\left(\frac{2\pi kn}{N}\right) + x_I(n) \sin\left(\frac{2\pi kn}{N}\right) \right]$$

$$X_I(k) = - \sum_{n=0}^{N-1} \left[ x_R(n) \sin\left(\frac{2\pi kn}{N}\right) - x_I(n) \cos\left(\frac{2\pi kn}{N}\right) \right]$$

Direct computation of above requires

- 1)  $2N^2$  evaluations of trigonometric functions
- 2)  $4N^2$  real multiplications ✓
- 3)  $4N(N-1)$  real additions.
- 4) A number of indexing and addressing operations.

## Radix - 2 FFT Algorithms

The fundamental principle of these algorithms is based on decomposing the computation of the DFT of a sequence of length  $N$  into successive smaller DFT's.

There are two basic classes of FFT algorithms :



1) Decimation in time (DIT)

2) Decimation in Frequency (DIF)

### 1. DIT FFT Algorithm to compute DFT of a Sequence

In DIT FFT, the  $N$ -point DFT is divided into two parts, one consisting of even numbered samples and the other consisting of odd numbered samples. The algorithm is developed assuming  $N$  is a power of 2.

The DFT of a sequence  $x(n)$  is given by

$$X(k) = \sum_{n=0}^{N-1} x(n) (W_N)^{kn} \quad \text{--- ①}$$

Separating  $x(n)$  into its even and odd numbered points.

$$X(k) = \sum_{n=0}^{N-2} x(n) (W_N)^{nk} + \sum_{n=1}^{N-1} x(n) (W_N)^{nk}$$

for  $n$ -even                      for  $n$ -odd

--- ②

substituting  $m = 2r$  for  $m$ -even &  
 $m = 2r+1$  for  $m$ -odd in eqn (2),

$$\left. \begin{array}{l} \text{ie for } m = 2r \\ \text{at } m=0, r=0 \\ \text{at } m=N-2, r = \frac{N}{2}-1 \end{array} \right\} \begin{array}{l} \text{for } m = 2r+1 \\ \text{at } m=0, r=0 \\ m=N-1, r = \frac{N}{2}-1 \end{array}$$

$\therefore$  Eqn (2) becomes,

$$X(k) = \sum_{r=0}^{\frac{N}{2}-1} x(2r) (W_N)^{2rk} + \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) (W_N)^{(2r+1)k} \quad \text{--- (3)}$$

In above eqn  $(W_N)^{2rk}$  can also be written as

$$\begin{aligned} (W_N)^{2rk} &= \left( e^{-j\frac{2\pi}{N}} \right)^{2rk} = \left( e^{-j\frac{2\pi}{N/2}} \right)^{rk} \\ &= (W_{N/2})^{rk} \end{aligned}$$

$$\therefore X(k) = \sum_{r=0}^{N/2-1} x(2r) (W_{N/2})^{rk} + W_N^k \sum_{r=0}^{N/2-1} x(2r+1) (W_{N/2})^{rk}$$

$$\boxed{X(k) = G_1(k) + W_N^k H(k)} \quad \text{--- (4)}$$

$\uparrow$   
 $N$ -POINT  
DFT

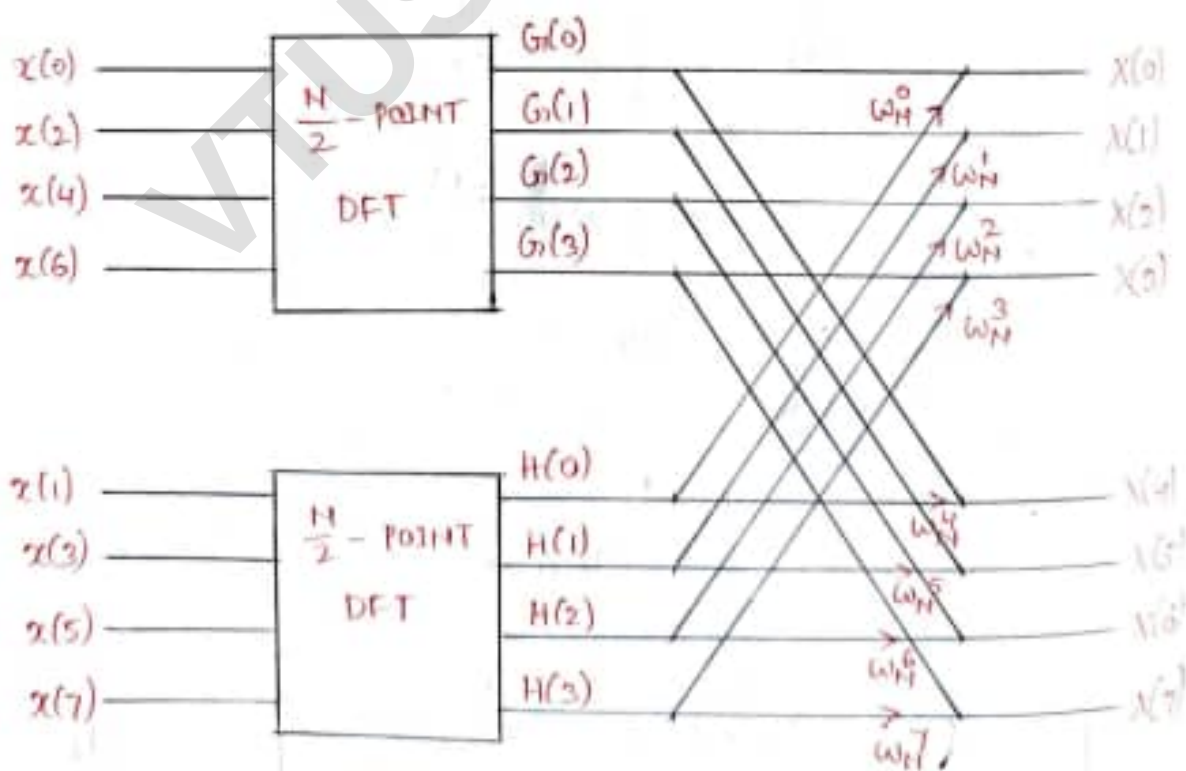
$\uparrow$   
 $\frac{N}{2}$ -POINT  
DFT

$\uparrow$   
 $\frac{N}{2}$ -POINT  
DFT

Where  $G(k) = \sum_{r=0}^{\frac{N}{2}-1} x(2r) (W_N/2)^{rk}$

$\Delta H(k) = \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) (W_N/2)^{rk}$

Each term in eqn (4) is an  $\frac{N}{2}$  point DFT  
 The first term  $[G(k)]$  represents  $\frac{N}{2}$  point DFT of even numbered points of the original sequence & the second sum  $[H(k)]$  is a  $\frac{N}{2}$  - point DFT of odd numbered points.



The values of  $x(0), x(1), \dots, x(7)$  can be evolved using eqn (4).

$$\text{ie, } x(0) = G(0) + W_N^0 H(0)$$

$$x(1) = G(1) + W_N^1 H(1)$$

$$x(2) = G(2) + W_N^2 H(2)$$

$$x(3) = G(3) + W_N^3 H(3)$$

$$x(4) = G(4) + W_N^4 H(4)$$

But from periodicity property,

$$\text{W.K.T } x(k) = x(k+N)$$

$$\text{ie, } G(4) = G(0+4) = G(0)$$

$$H(4) = H(0+4) = H(0)$$

$$\therefore x(4) = G(0) + W_N^4 H(0)$$

$$\text{Similarly } x(5) = G(1) + W_N^5 H(1)$$

$$x(6) = G(2) + W_N^6 H(2)$$

$$x(7) = G(3) + W_N^7 H(3)$$

Since  $\frac{N}{2}$  is even, we can divide

$\frac{N}{2}$ -point DFT into two  $\frac{N}{4}$ -point

DFTs which combine to form a

$\frac{N}{2}$  point DFT.

From Eqn (7) consider  $G(k)$

$$\text{ie, } G(k) = \sum_{r=0}^{\frac{N}{2}-1} x(2r) (W_{N/2})^{rk}$$

separating  $x(2r)$  into even and odd numbered points.

$$G(k) = \underbrace{\sum_{r=0}^{\frac{N}{2}-2} x(2r) (W_{N/2})^{rk}}_{\text{Even}} + \underbrace{\sum_{r=1}^{\frac{N}{2}-1} x(2r) (W_{N/2})^{rk}}_{\text{Odd}}$$

Put  $r = 2l$  for even

at  $r=0$ ,  $l=0$

$$r = \frac{N}{2} - 2, \quad l = \frac{N}{4} - 1$$

Put  $r = 2l + 1$  for odd

at  $r=1$ ,  $l=0$

$$r = \frac{N}{2} - 1, \quad \frac{N}{2} - 1 = 2l + 1 \quad \text{or} \quad 2l = \frac{N}{2} - 2$$

$$l = \frac{N}{4} - 1$$

$$\therefore G(k) = \sum_{l=0}^{\frac{N}{4}-1} x(4l) (W_{N/2})^{2lk} + \sum_{l=0}^{\frac{N}{4}-1} x(4l+2) (W_{N/2})^{(2l+1)k}$$



$$= \sum_{l=0}^{\frac{N}{4}-1} x(4l) (W_{N/4})^{lk} + (W_{N/2})^k \sum_{l=0}^{\frac{N}{4}-1} x(4l+2) (W_{N/4})^{lk}$$

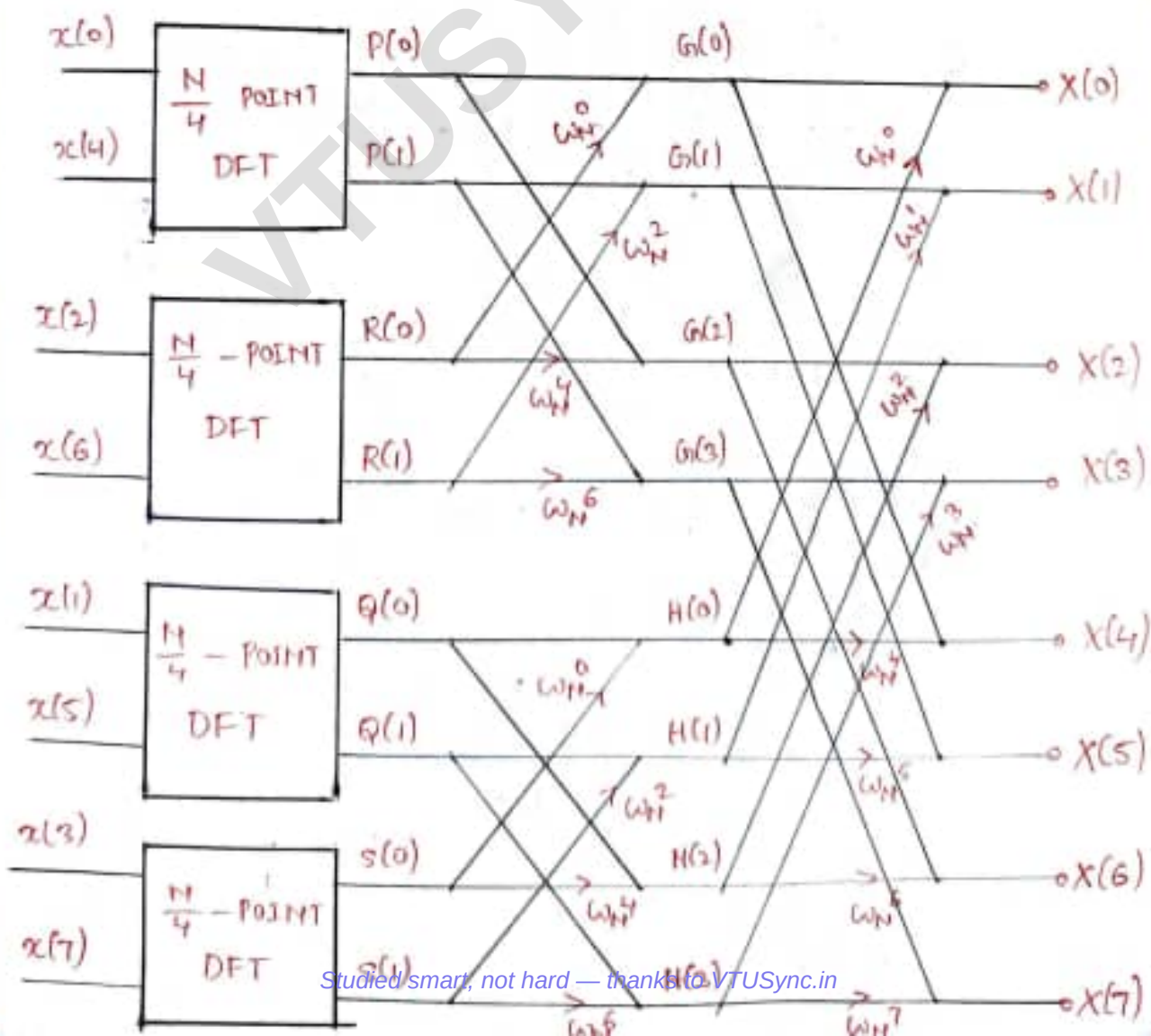
or  $G(k) = P(k) + (W_{N/2})^k R(k)$

or  $G(k) = P(k) + (W_N)^{2k} R(k)$

$\uparrow$                        $\uparrow$                        $\uparrow$   
 $\frac{N}{2}$  - PT DFT       $\frac{N}{4}$  - PT DFT       $\frac{N}{4}$  - PT DFT

where  $P(k) = \sum_{l=0}^{\frac{N}{4}-1} x(4l) (W_{N/4})^{lk}$

$\Delta$   $R(k) = \sum_{l=0}^{\frac{N}{4}-1} x(4l+2) (W_{N/4})^{lk}$





III) consider  $H(k)$  and can be simplified as

$$H(k) = Q(k) + W_N^{2k} \cdot S(k)$$

$\uparrow$                        $\uparrow$                        $\uparrow$   
 $\frac{N}{2}$ -PT DFT       $\frac{N}{4}$ -PT DFT       $\frac{N}{4}$ -PT DFT

using  $G(k)$  &  $H(k)$  above Butterfly structure can be drawn.

The final step in the computation of 8-pt DFT by DIT-FFT algorithm is the computation of 2-PT DFT.

Consider  $P(k)$

$$P(k) = \sum_{l=0}^{\frac{N}{4}-1} x(4l) \cdot (W_{N/4})^{lk}$$

for  $N=8$

$$P(k) = \sum_{l=0}^1 x(4l) (W_N)^{4lk}$$

$$P(k) = x(0) W_N^0 + x(4) (W_N)^{4k}$$

$$W_N^0 = 1$$

$$W_N^{4k} = e^{-j \frac{2\pi}{8} \cdot 4k}$$

$$= e^{-j\pi k}$$

$$\therefore W_N^{4k} = (-1)^k$$

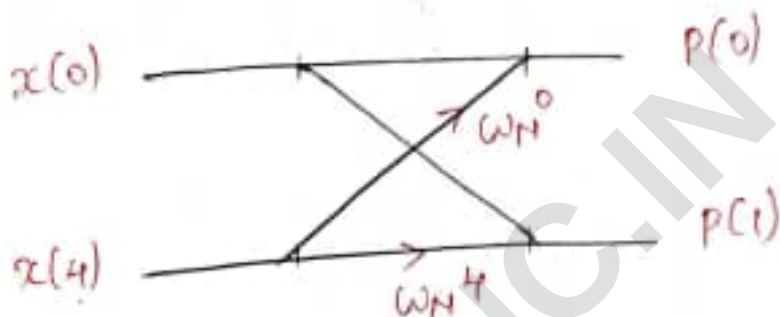
at  $k=0$ ,

$$p(0) = x(0) + w_N^0 \cdot x(4)$$

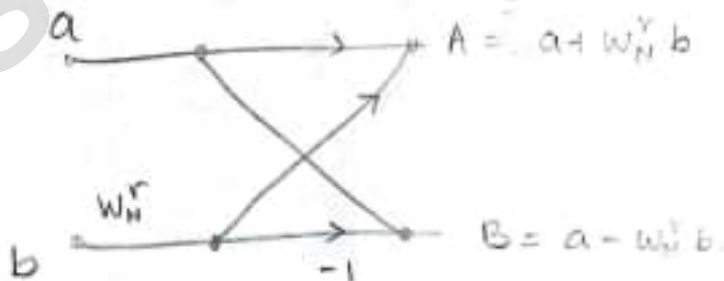
$k=1$ ,

$$p(1) = x(0) + (w_N)^4 \cdot x(4)$$

Using this each  $\frac{N}{4}$ -PT DFT can be written as



Basic butterfly  
computation in  
DFT - FFT



(P.T.O)

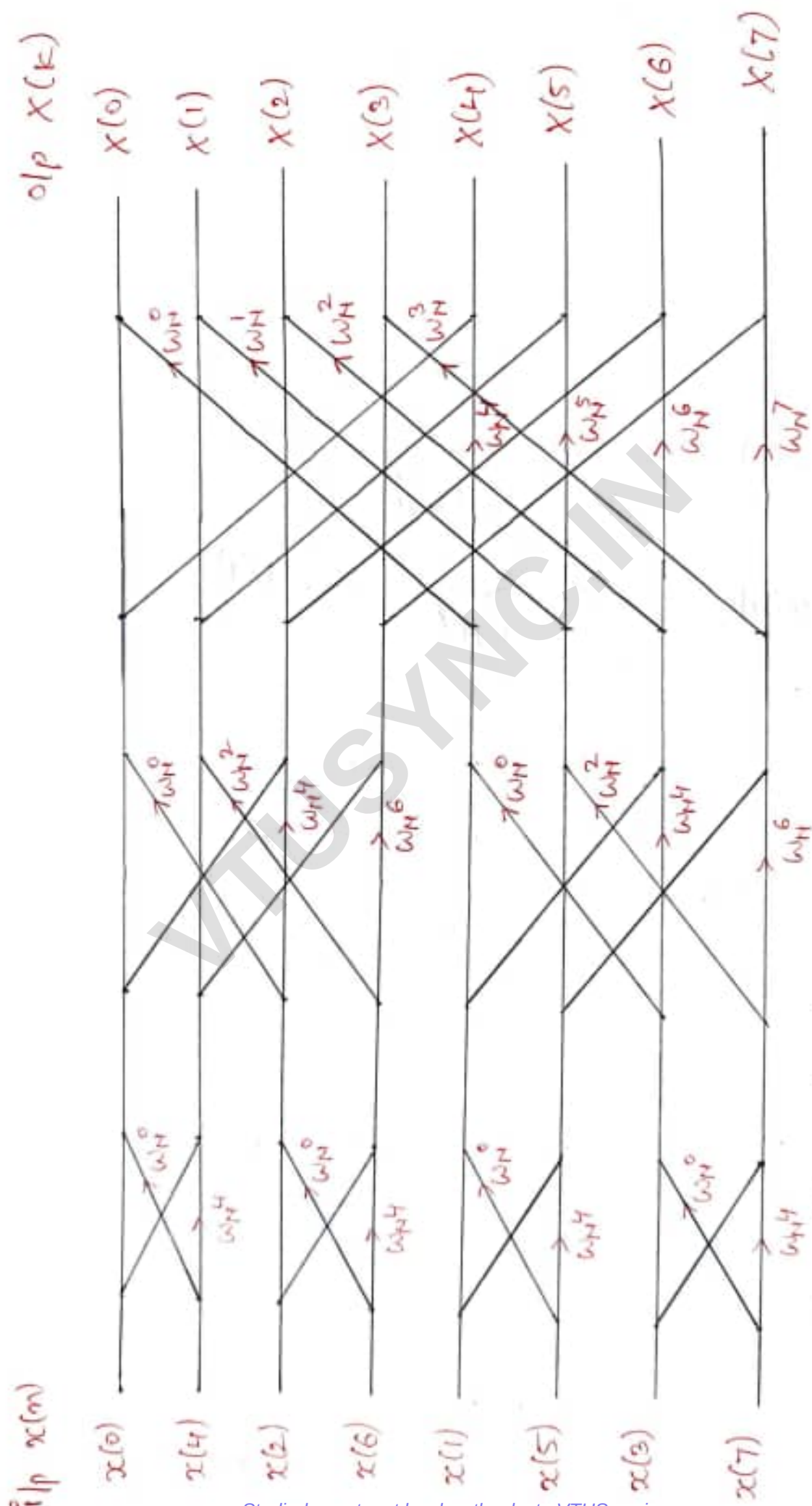
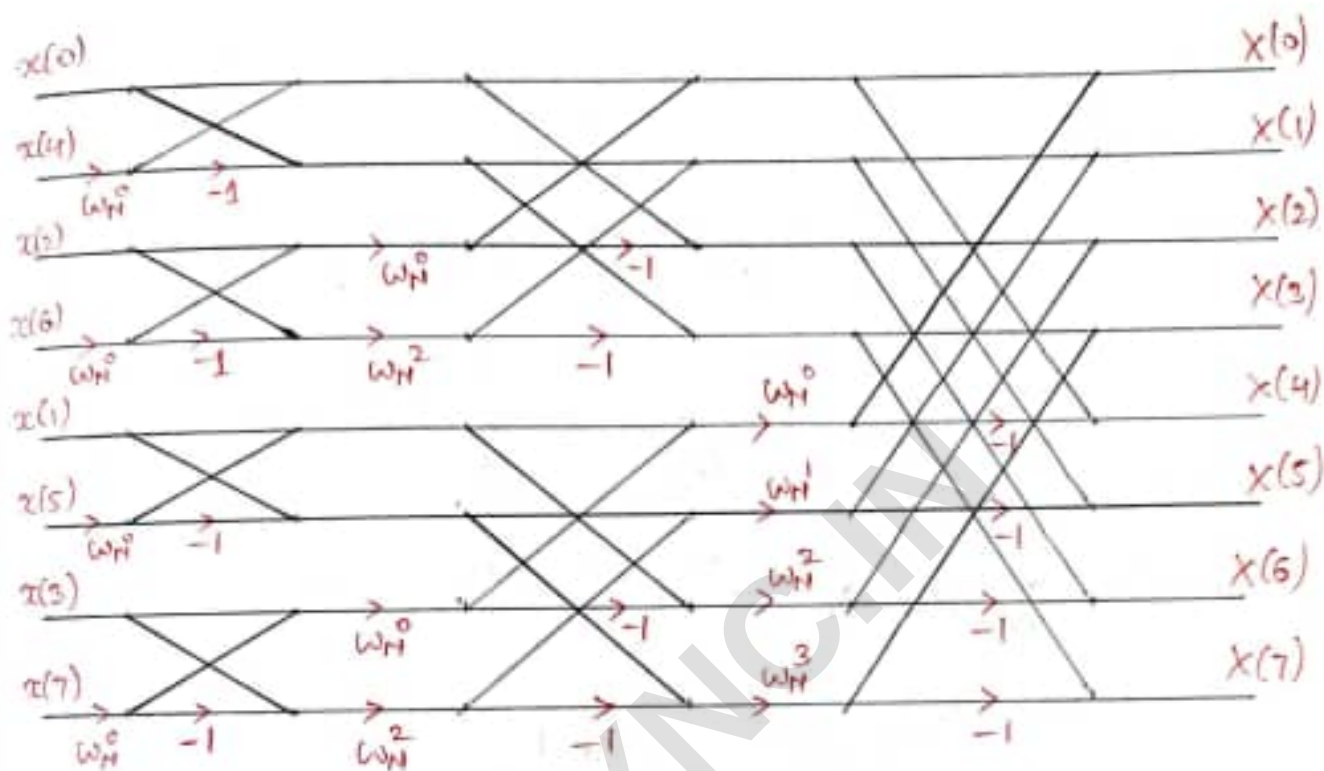


Fig. Complete signal flow graph for  $N=8$ . 3-stages with  $N$  multiplications for each stage.

Using symmetric property we can simplify above signal flow graph as



From the above signal flow graph we observe that, there are 3-stages and each stage requires 4 multiplications.

$$\therefore \text{No. of complex multiplications} = \frac{\text{No. of multiplications per stage} \times \text{No. of stages}}{1}$$

$$= \frac{N}{2} \cdot \log_2 N$$

$$\text{No. of complex Additions} = N \cdot \log_2 N$$



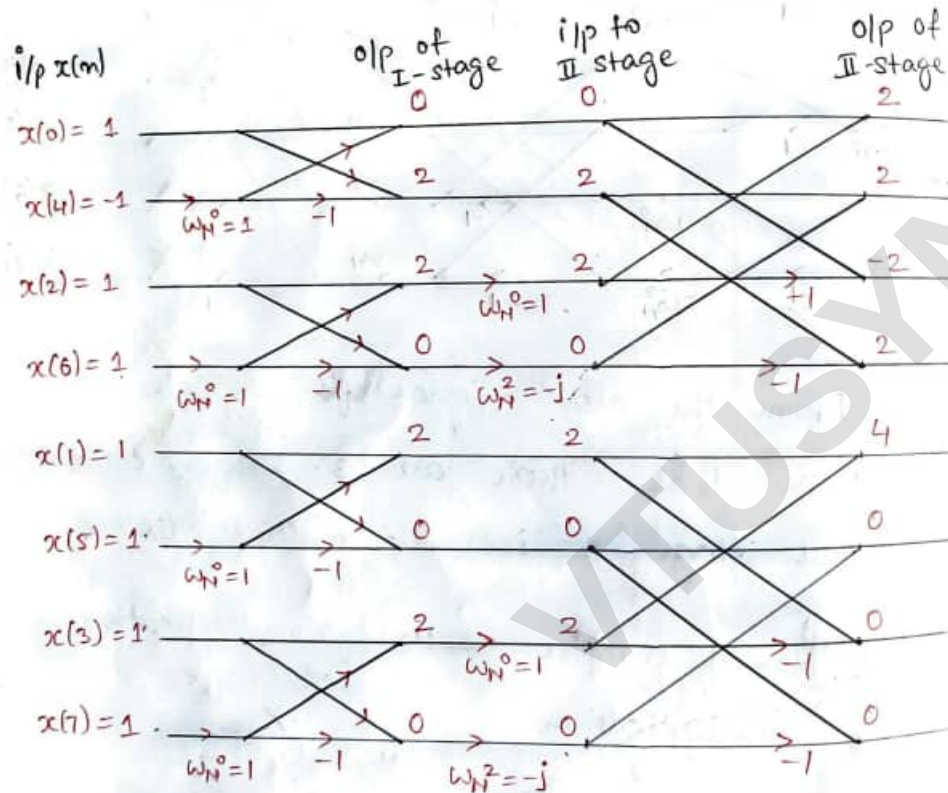
⑦ Compute the DFT of a sequence using DIT FFT algorithm  $x(m) = \{1, 1, 1, 1, 1, -1, 1, 1\}$

Soln:-  $N=8$

$$W_8^0 = 1, \quad W_8^1 = e^{-j\frac{2\pi}{8}} = 0.707 - j0.707$$

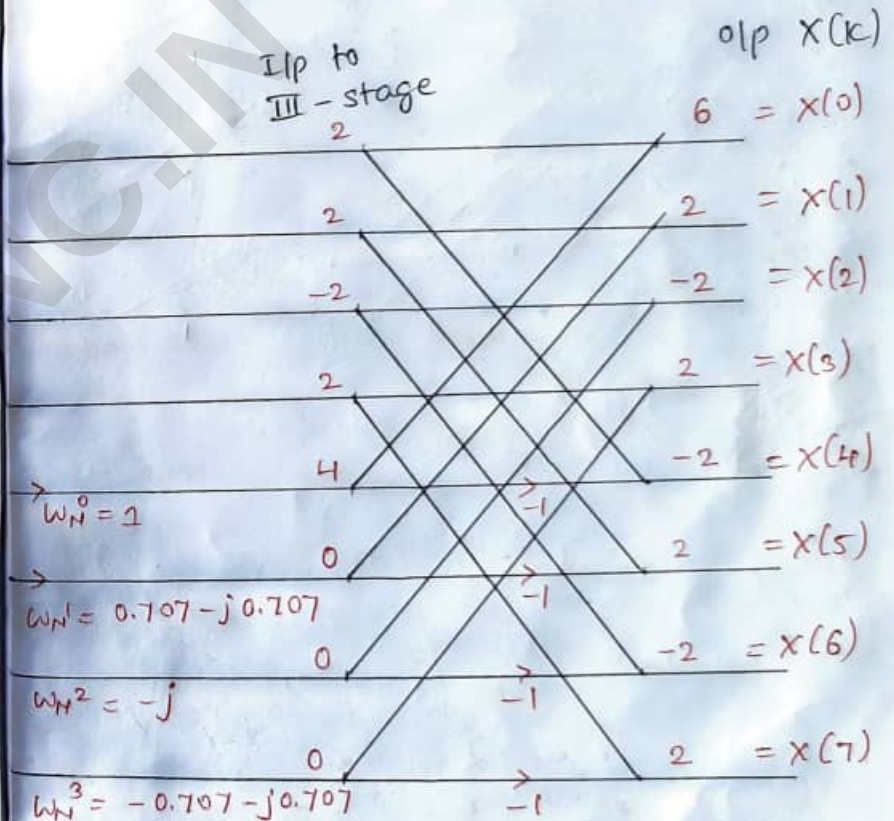
$$W_8^2 = e^{-j\frac{2\pi}{8} \times 2} = -j$$

$$W_8^3 = e^{-j\frac{2\pi}{8} \times 3} = -0.707 - j0.707$$



$\therefore$  DFT of sequence  $x(m)$  is

$$X(k) = \{6, 2, -2, 2, -2, 2, -2, 2\}$$



## 2) DIF FFT Algorithm to compute DFT of a sequence

DIF-FFT algorithm is based on the decomposition of  $N$ -point DFT coefficients  $X(k)$  into smaller and smaller subsequences.

Consider  $N$ -point DFT sequence  $x(n)$

$$X(k) = \sum_{n=0}^{N-1} x(n) (W_N)^{nk} \quad \text{--- (1)}$$

we can ~~divide~~ divide the  $N$ -point sequence into the first half and last half of the points.

$$X(k) = \sum_{n=0}^{\frac{N}{2}-1} x(n) (W_N)^{nk} + \sum_{n=\frac{N}{2}}^{N-1} x(n) (W_N)^{nk} \quad \text{--- (2)}$$

Put  $m = n - \frac{N}{2}$  in the second summation

$$\text{at } n = \frac{N}{2}, \quad m = \frac{N}{2} - \frac{N}{2} = 0$$

$$n = N-1, \quad m = N-1 - \frac{N}{2} = \frac{N}{2} - 1$$

$$\text{and } n = m + \frac{N}{2}$$

$$\therefore X(k) = \sum_{n=0}^{\frac{N}{2}-1} x(n) (W_N)^{nk} + \sum_{m=0}^{\frac{N}{2}-1} x\left(m + \frac{N}{2}\right) (W_N)^{\left(m + \frac{N}{2}\right)k}$$



Replace  $m$  by  $m$  in II summation

$$X(k) = \sum_{n=0}^{\frac{N}{2}-1} x(n) W_N^{nk} + \sum_{n=0}^{\frac{N}{2}-1} x(n+\frac{N}{2}) (W_N)^{(n+\frac{N}{2})k}$$

consider  $(W_N)^{(n+\frac{N}{2})k} = W_N^{nk} \cdot W_N^{N/2 \cdot k}$

$$= W_N^{nk} \cdot e^{-j \frac{2\pi}{N} \times \frac{N}{2} \times k}$$

$$= (-1)^k W_N^{nk}$$

$$\therefore X(k) = \sum_{n=0}^{\frac{N}{2}-1} \left[ x(n) + (-1)^k x(n+\frac{N}{2}) \right] W_N^{nk} \quad \text{--- (3)}$$

Consider even and odd parts of  $X(k)$   
 ie, put  $k=2r$  for even numbered points  
 $\angle k=2r+1$  for odd numbered points

$$\therefore X(2r) = \sum_{n=0}^{\frac{N}{2}-1} \left[ x(n) + (-1)^{2r} x(n+\frac{N}{2}) \right] W_N^{2rm}$$

$$X(2r+1) = \sum_{n=0}^{\frac{N}{2}-1} \left[ x(n) + (-1)^{2r+1} x(n+\frac{N}{2}) \right] W_N^{(2r+1)m}$$

consider  $(W_N)^{(2r+1)m} = (W_N)^{2rm} \cdot (W_N)^m$

$$= (W_{N/2})^{rm} \cdot (W_N)^m$$

$$\therefore X(2r) = \sum_{m=0}^{\frac{N}{2}-1} \left[ x(m) + x\left(m + \frac{N}{2}\right) \right] (W_{N/2})^{rm} \quad \text{--- (4)}$$

$$X(2r+1) = \sum_{m=0}^{\frac{N}{2}-1} \left[ x(m) - x\left(m + \frac{N}{2}\right) \right] W_N^m \cdot (W_{N/2})^{rm} \quad \text{--- (5)}$$

$$\therefore (-1)^{2r} = 1$$

$$\& (-1)^{2r+1} = -1$$

$$\text{Let } g(m) = x(m) + x\left(m + \frac{N}{2}\right)$$

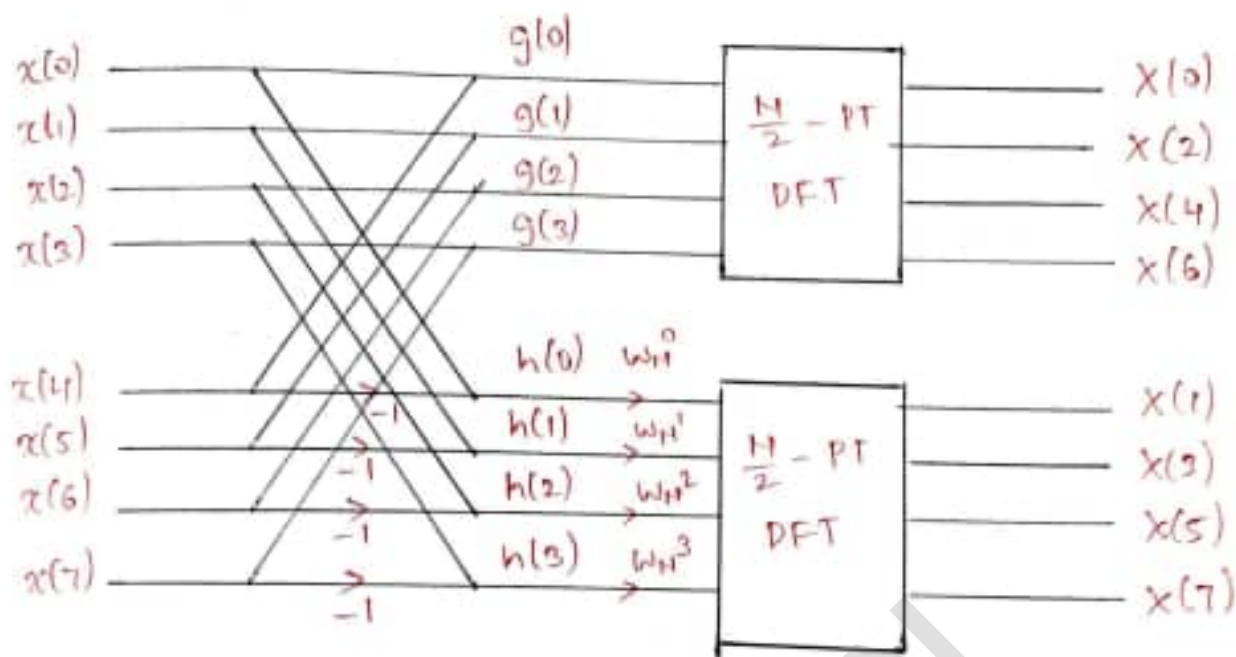
$$\& h(m) = x(m) - x\left(m + \frac{N}{2}\right)$$

Hence eqn (4) & (5) can be written as

$$X(2r) = \sum_{m=0}^{\frac{N}{2}-1} g(m) (W_{N/2})^{rm} \quad \text{--- (6)}$$

$$X(2r+1) = \sum_{m=0}^{\frac{N}{2}-1} \left[ h(m) W_N^m \right] (W_{N/2})^{rm} \quad \text{--- (7)}$$

Using eqn's (6) & (7)  $\frac{N}{2}$  - point DFT can be written as



$\frac{N}{4}$  - POINT DFT :-

Consider Eqn (6)

$$X(2r) = \sum_{n=0}^{\frac{N}{2}-1} g(n) (w_{N/2})^{rn}$$

Dividing above summation into two equal halves and solving for

$r = 2l$  for even and  $r = 2l+1$  for

odd we get

$$X(4l) = \sum_{n=0}^{\frac{N}{4}-1} p(n) (w_{N/4})^{ln}$$

$$X(4l+2) = \sum_{n=0}^{\frac{N}{4}-1} [Q(n) (w_N)^{2n}] (w_{N/4})^{ln}$$

where  $p(n) = g(n) + g(n + \frac{N}{4})$

&  $Q(n) = g(n) - g(n + \frac{N}{4})$

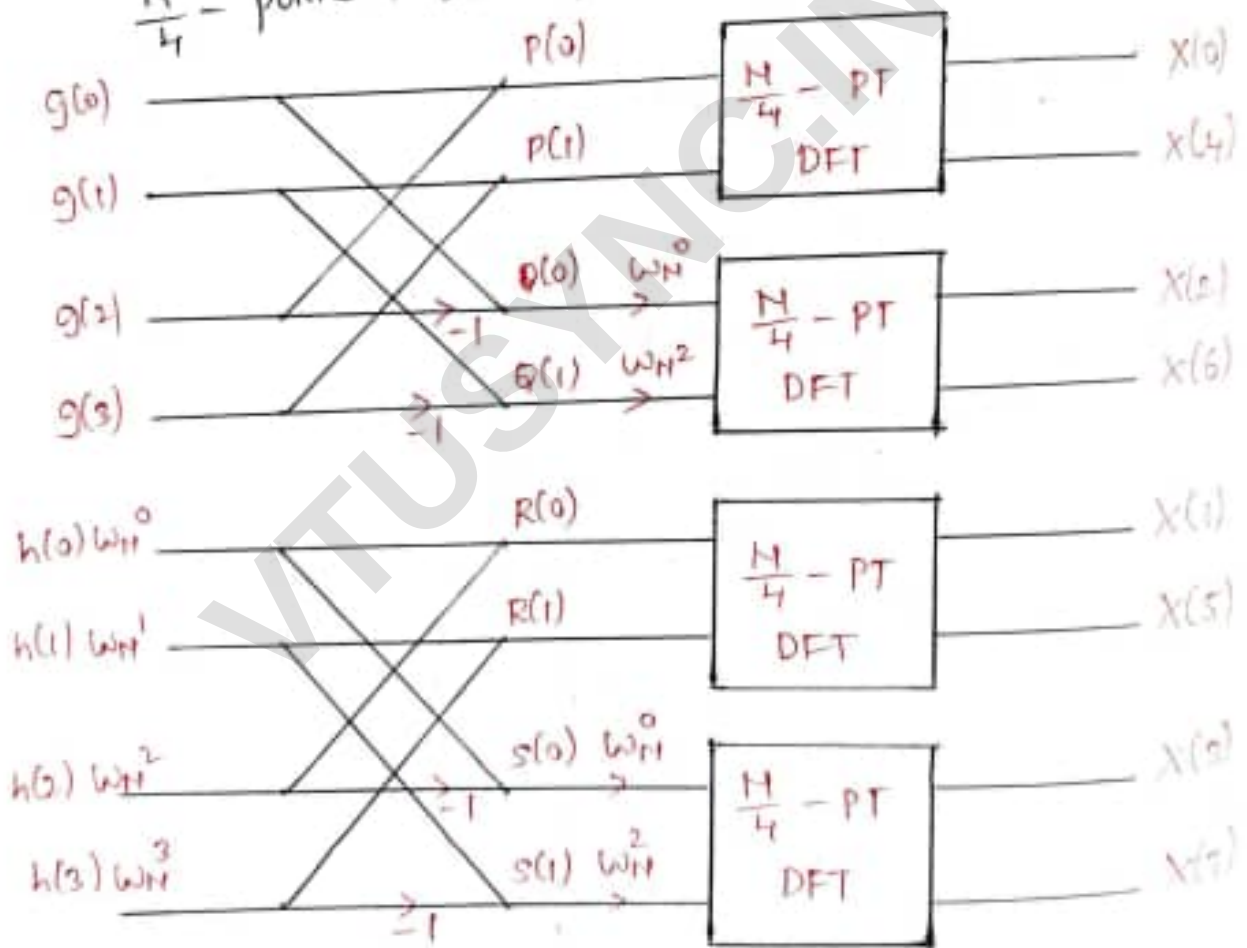
||| by consider eqn (7) and solving we get

$$X(4l+1) = \sum_{n=0}^{\frac{N}{4}-1} R(n) (W_{N/4})^{ln}$$

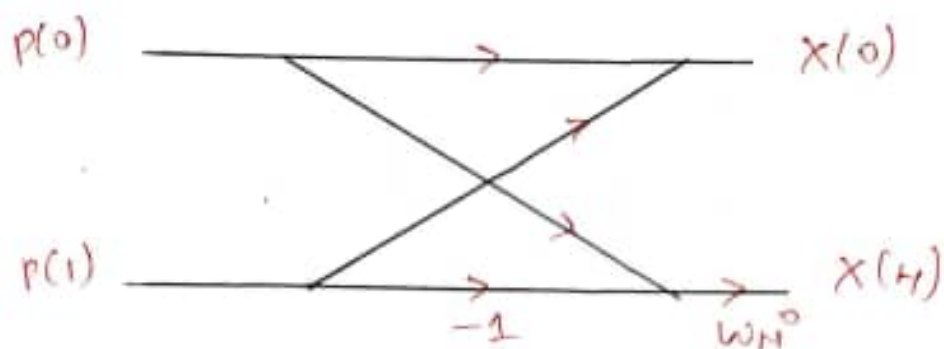
$$X(4l+3) = \sum_{n=0}^{\frac{N}{4}-1} [S(n) W_N^{2n}] (W_{N/4})^{ln}$$

using above all eqn's we can draw

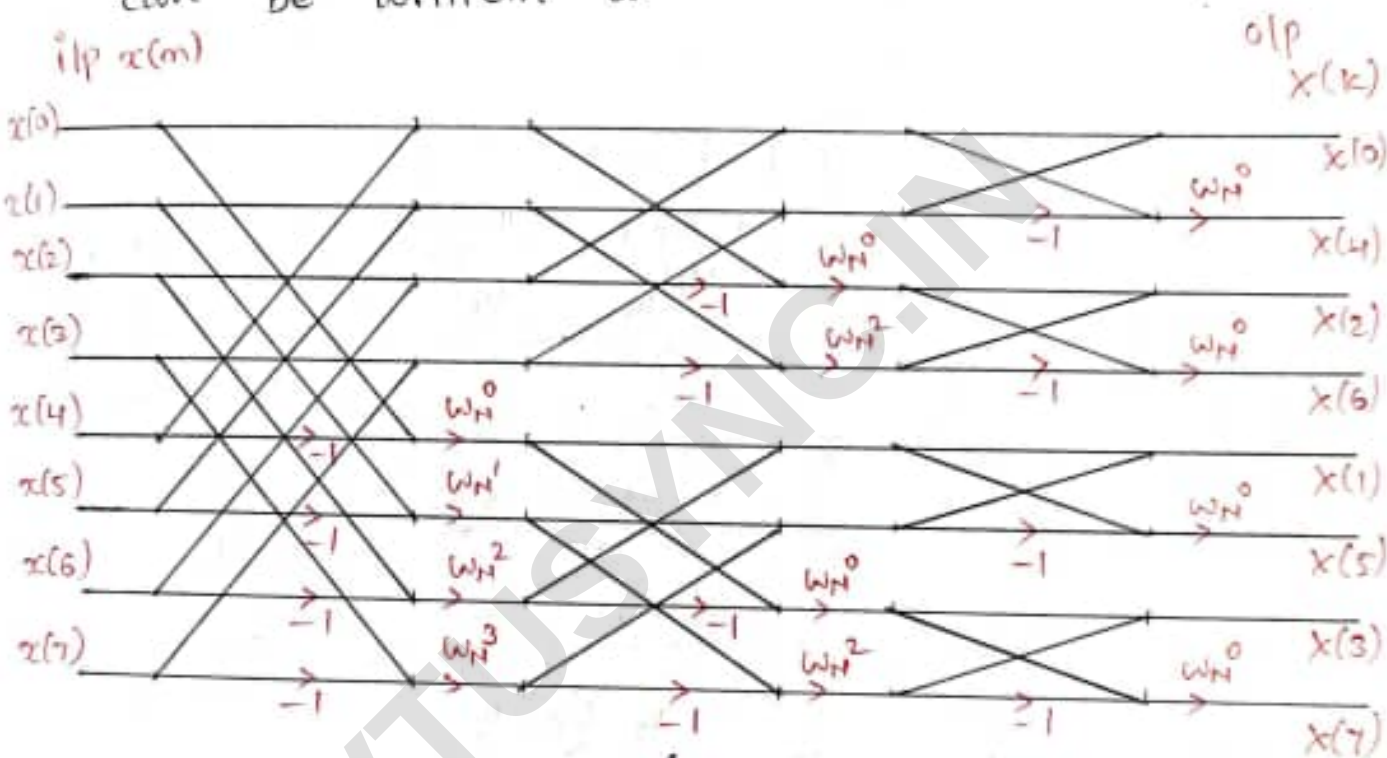
$\frac{N}{4}$  - point DFT as



Two point DFT can be easily implemented by taking the sum and difference of the last stage i/p's.



For  $N=8$  Complete signal flow graph can be written as



Since  $N=8$  It requires  $\log_2 8 = 3$  stages  
each stage require  $\frac{N}{2}$  complex multipli-  
cations &  $N$ - complex additions.

$\therefore$  We require  $\frac{N}{2} \log_2 N$  complex multiplica-  
tions.

&  $N \cdot \log_2 N$  complex additions.



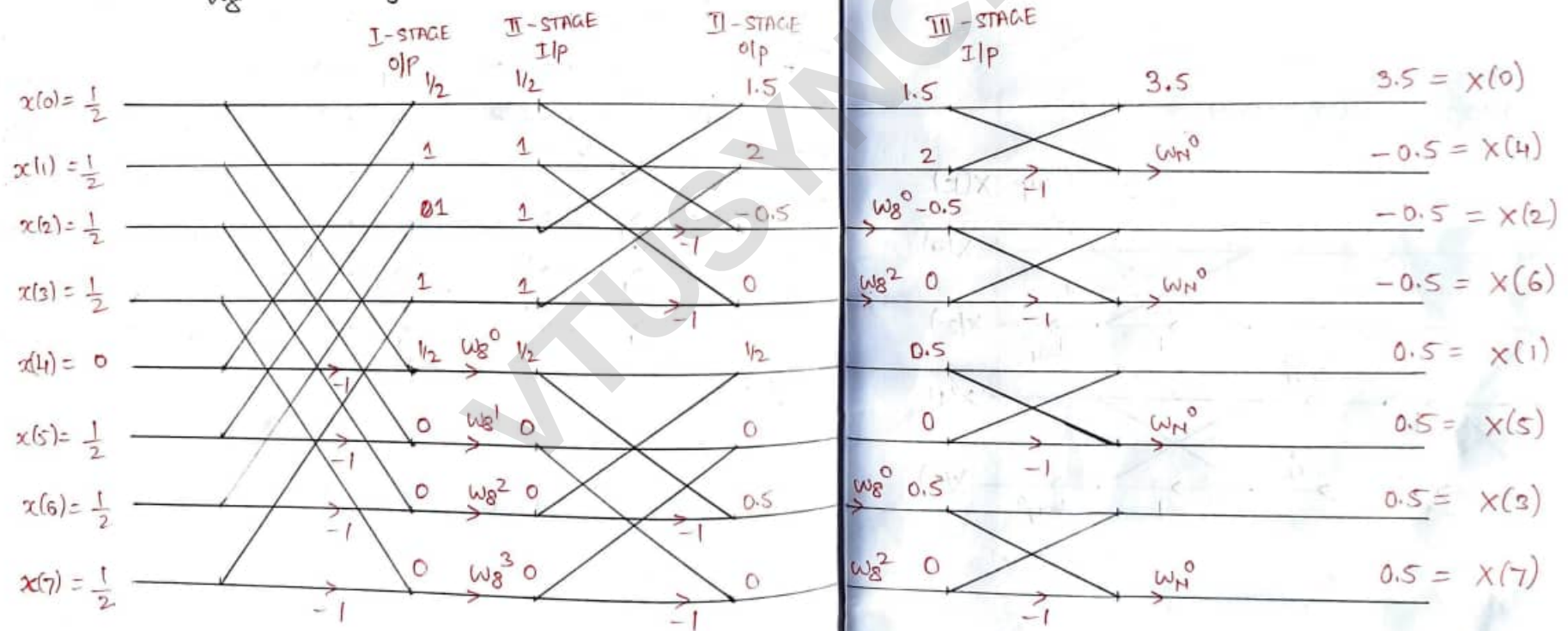
⑦ Compute the DFT of the sequence using DIF-FFT algorithm.  
 $x(m) = \{ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \}$

Soln: -  $N=8$

$$\therefore W_8^0 = 1$$

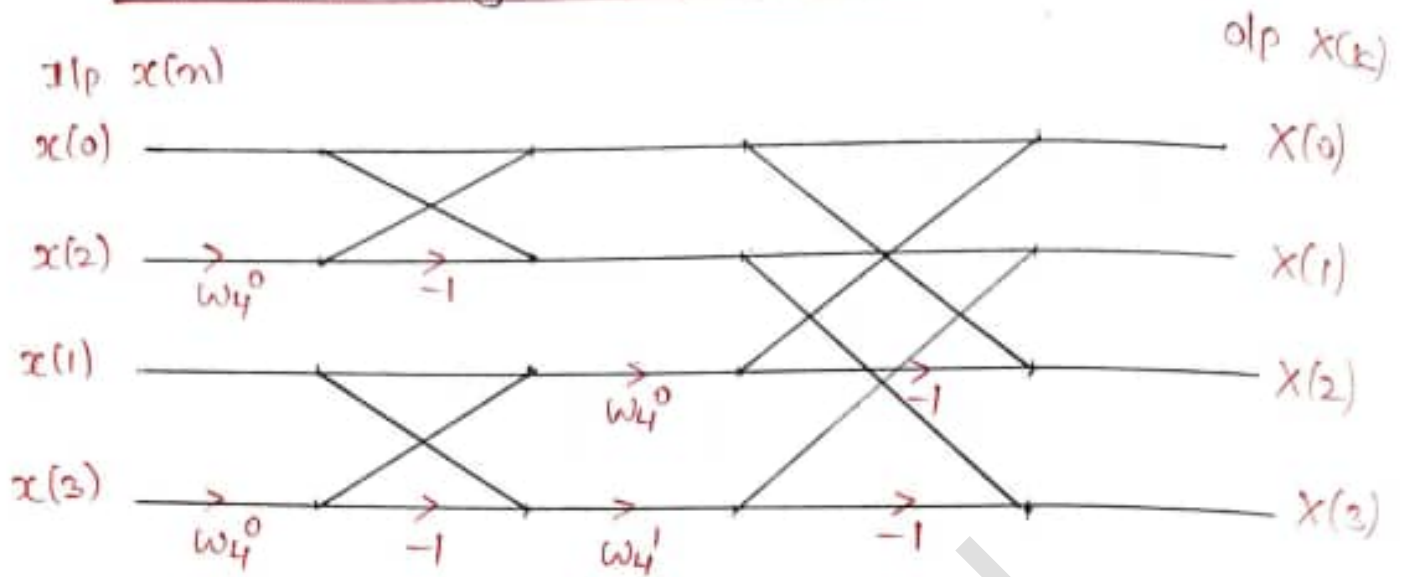
$$W_8^2 = -j$$

$$W_8^1 = 0.707 - j0.707 \quad W_8^3 = -0.707 - j0.707$$

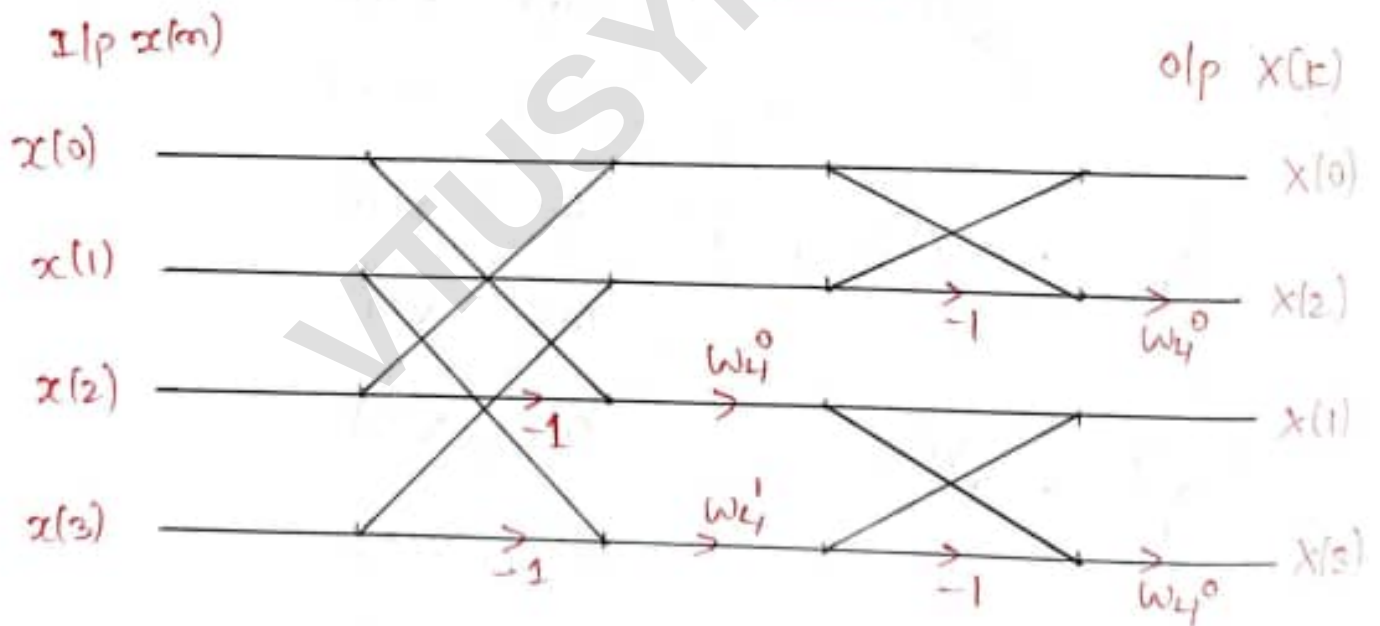


$$\therefore X(k) = \{ 3.5, +0.5, -0.5, +0.5, -0.5, 0.5, -0.5, 0.5 \}$$

## DIT-FFT Diagram for $N=4$



## DIF-FFT Diagram for $N=4$



Find the DFT of following sequence using DIT/DIF FFT algorithms.

$$1) x(m) = \{ 2, 2, 2, 2, 1, 1, 1, 1 \}$$

$$\text{Soln: } X(k) = \{ 12, 1 - j2.414, 0, 1 - j0.414, 0, 1 + j0.414, 0, 1 + j2.414 \}$$

$$2) x(m) = \{ 1, 2, 3, 4, 4, 3, 2, 1 \}$$

$$\text{Soln: } X(k) = \{ 20, -5.828 - j2.414, 0, -0.17157 - j0.4142, 0, -0.17157 + j0.4142, 0, -5.82843 + j2.4142 \}$$

$$3) x(m) = \{ 1, 1, 1, 1, 1, 1, 1, 1 \}$$

$$X(k) = \{ 8, 0, 0, 0, 0, 0, 0, 0 \}$$

$$4) x(m) = \{ 0.5, 0.5, 0.5, 0.5, 0, 0, 0, 0 \}$$

$$X(k) = \{ 2, (0.5 - j1.207), 0, (0.5 - j0.207), 0, (0.5 + j0.207), 0, (0.5 + j1.207) \}$$

$$5) \quad x(m) = \left\{ \frac{1}{\sqrt{2}}, 1, \frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}, -1, -\frac{1}{\sqrt{2}}, 0 \right\}$$

$$X(k) = \left\{ 0, 2\sqrt{2} - j2\sqrt{2}, 0, 0, 0, 0, 0, 2\sqrt{2} + j2\sqrt{2} \right\}$$

$$6) \quad x(m) = m+1, \quad 0 \leq m \leq 7$$

$$\text{ie } x(m) = \{ 1, 2, 3, 4, 5, 6, 7, 8 \}$$

$$X(k) = \left\{ 36, (-4 + j9.6), (-4 + 4j), (-4 + 1.6j), \right. \\ \left. -4, (-4 - 1.6j), (-4 - 4j), (-4 - 9.6j) \right\}$$

$$7) \quad x(m) = \cos\left(\frac{m\pi}{2}\right) \quad \text{for } N=4$$

$$\therefore x(m) = \{ 1, 0, -1, 0 \}$$

$$X(k) = \{ 0, 2, 0, 2 \}$$

$$8) \quad x(m) = \{ 0.5, 0.5, 0.5, 0.5, 0, 0, 0, 0 \}$$

$$X(k) = \left\{ 2, (0.5 - j1.207), 0, (0.5 - j0.207), 0, \right. \\ \left. (0.5 + j0.207), 0, (0.5 + j1.207) \right\}$$

$$9) \quad x(m) = \sin\left(\frac{\pi}{2}m\right) \quad 0 \leq m \leq 7$$

$$x(m) = \{ 0, 0, -j4, 0, 0, 0, j4, 0 \}$$

## DIT-FFT Algorithm to compute IDFT of a sequence

The IDFT of an  $N$ -point sequence  $X(k)$  is given by,

$$x(m) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) (W_N)^{-km} \quad \text{--- (1)}$$

Using DIT FFT algorithm to compute IDFT involves the decomposition of  $x(m)$  into smaller subsequences. Dividing above summation into two equal halves.

$$x(m) = \frac{1}{N} \left[ \sum_{k=0}^{\frac{N}{2}-1} X(k) (W_N)^{-km} + \sum_{k=\frac{N}{2}}^{N-1} X(k) (W_N)^{-km} \right]$$

$$\text{Put } k - \frac{N}{2} = m \Rightarrow k = m + \frac{N}{2}$$

$$\text{at } k = \frac{N}{2} \quad m = 0$$

$$k = N-1 \quad m = \frac{N}{2} - 1$$

$$x(m) = \frac{1}{N} \left[ \sum_{k=0}^{\frac{N}{2}-1} X(k) (W_N)^{-km} + \sum_{m=0}^{\frac{N}{2}-1} X(m + \frac{N}{2}) (W_N)^{-km} \right]$$

Replace  $m$  by  $k$  in 2nd summation.

$$x(m) = \frac{1}{N} \left[ \sum_{k=0}^{\frac{N}{2}-1} X(k) (W_N)^{-km} + \sum_{k=0}^{\frac{N}{2}-1} X(k + \frac{N}{2}) (W_N)^{-km} \right]$$



$$= \frac{1}{N} \sum_{k=0}^{\frac{N}{2}-1} \left[ x(k) + x\left(k + \frac{N}{2}\right) W_N^{-mN/2} \right] W_N^{-km}$$

$$W_N^{-mN/2} = \left( e^{-j\frac{2\pi}{N}} \right)^{-\frac{mN}{2}} = e^{+jm\pi}$$

$$= (-1)^m$$

$$x(m) = \frac{1}{N} \sum_{k=0}^{\frac{N}{2}-1} \left[ x(k) + (-1)^m x\left(k + \frac{N}{2}\right) \right] W_N^{-km} \quad \text{--- (2)}$$

Dividing  $x(m)$  into even and odd components put  $m = 2r$  for even and  $m = 2r+1$  for odd.

$$x(2r) = \frac{1}{N} \sum_{k=0}^{\frac{N}{2}-1} \left[ x(k) + x\left(k + \frac{N}{2}\right) \right] W_N^{-k2r} \quad \text{--- (3)}$$

$$x(2r+1) = \frac{1}{N} \sum_{k=0}^{\frac{N}{2}-1} \left[ x(k) - x\left(k + \frac{N}{2}\right) \right] W_N^{-k2r} \cdot W_N^{-k} \quad \text{--- (4)}$$

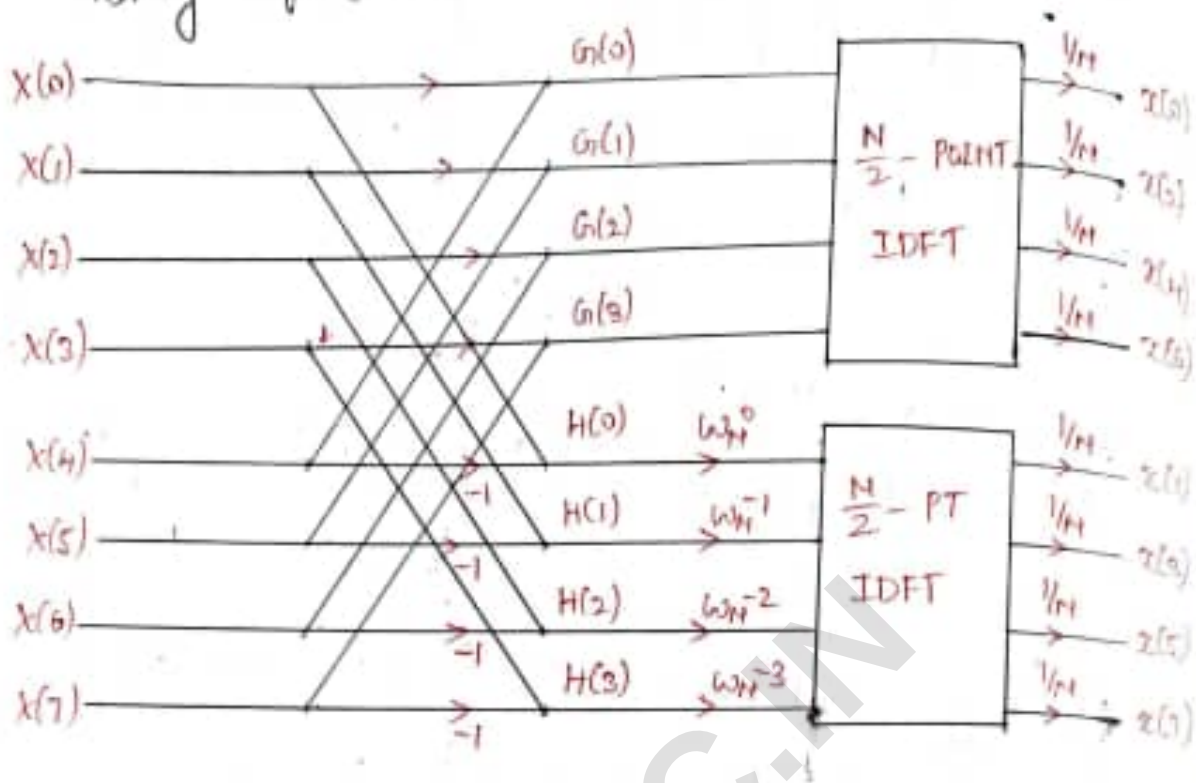
put  $G(k) = x(k) + x\left(k + \frac{N}{2}\right)$

$H(k) = x(k) - x\left(k + \frac{N}{2}\right)$

$$\therefore x(2r) = \frac{1}{N} \sum_{k=0}^{\frac{N}{2}-1} G(k) W_N^{-kr} \quad \text{--- (5)}$$

$$x(2r+1) = \frac{1}{N} \sum_{k=0}^{\frac{N}{2}-1} \left[ H(k) W_N^{-k} \right] W_N^{-kr} \quad \text{--- (6)}$$

using Eqns ⑤ & ⑥, we can sketch  $N/2$  point.



From Eqn ⑤,

$$x(2r) = \frac{1}{N} \sum_{k=0}^{N/2-1} G(k) W_N^{-kr}$$

Replace  $2r$  by  $r$  and dividing into two parts.

$$x(r) = \frac{1}{N} \left[ \sum_{k=0}^{N/4-1} G(k) W_N^{-kr} + \sum_{k=N/4}^{N/2-1} G(k) W_N^{-kr} \right]$$

put  $m = k - \frac{N}{4}$  at  $k = \frac{N}{4}$

at  $k = \frac{N}{4}$ ,  $m = 0$

$$k = \frac{N}{2} - 1 \quad m = \frac{N}{4} - 1 \quad \therefore$$

$$x(r) = \frac{1}{N} \left[ \sum_{k=0}^{N/4-1} G(k) W_N^{-kr} + \sum_{m=0}^{N/4-1} G(m + \frac{N}{4}) W_N^{-r(m + \frac{N}{4})} \right]$$

change  $m$  to  $k$  in II summation.

$$= \frac{1}{N} \left[ \sum_{k=0}^{N/4-1} G(k) W_{N/2}^{-kr} + \sum_{k=0}^{N/4-1} G(k + \frac{N}{4}) W_{N/2}^{-rk - \frac{rN}{4}} \right]$$

$$W_{N/2}^{-\frac{rN}{4}} = \left( e^{j \frac{2\pi r}{N/2}} \right)^{N/4} = (-1)^r$$

$$x(r) = \frac{1}{N} \left\{ \sum_{k=0}^{N/4-1} \left[ G(k) + (-1)^r G(k + \frac{N}{4}) \right] W_{N/2}^{-rk} \right\}$$

put  $r = 2l$  for even part and  $r = 2l+1$  for odd part.

$$x(2l) = \frac{1}{N} \left\{ \sum_{k=0}^{N/4-1} \left[ G(k) + G(k + \frac{N}{4}) \right] W_{N/2}^{-2lk} \right\}$$

$$x(2l+1) = \frac{1}{N} \left\{ \sum_{k=0}^{N/4-1} \left[ G(k) - G(k + \frac{N}{4}) \right] W_{N/2}^{-(2l+1)k} \right\}$$

$$\text{Let } P(k) = G(k) + G(k + \frac{N}{4})$$

$$Q(k) = G(k) - G(k + \frac{N}{4})$$

$$W_{N/2}^{-(2l+1)k} = W_{N/2}^{-2lk} W_{N/2}^{-k}$$

$$\therefore x(2l) = \frac{1}{N} \left\{ \sum_{k=0}^{N/4-1} P(k) W_{N/4}^{-lk} \right\} \quad \text{--- (7)}$$

$$x(2l+1) = \frac{1}{N} \left\{ \sum_{k=0}^{N/4-1} \left[ Q(k) W_{N/2}^{-k} \right] W_{N/4}^{-lk} \right\} \quad \text{--- (8)}$$

IIIly from (6), dividing into 2 parts & solving for even and odd parts.

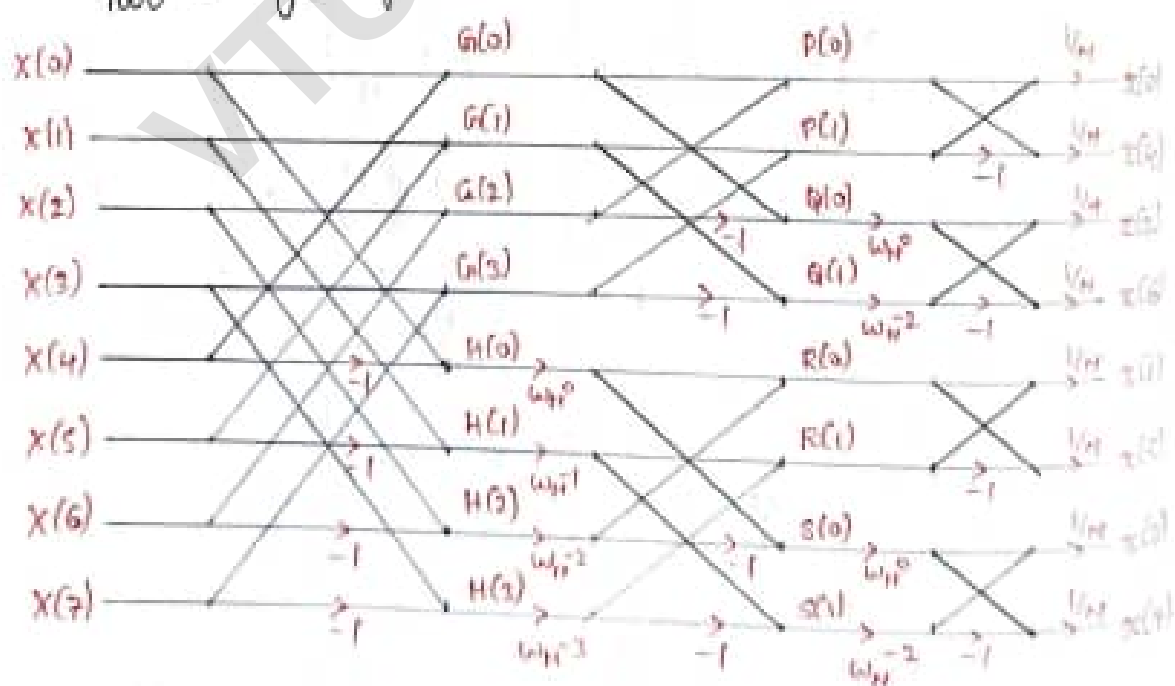
$$x(2l) = \frac{1}{N} \left\{ \sum_{k=0}^{\frac{N}{4}-1} R(k) W_{N/4}^{-lk} \right\}$$

$$x(2l+1) = \frac{1}{N} \left\{ \sum_{k=0}^{\frac{N}{4}-1} S(k) W_{N/4}^{-lk} \right\}$$

$$\text{where } R(k) = H(k) + H\left(k + \frac{N}{4}\right)$$

$$S(k) = H(k) - H\left(k + \frac{N}{4}\right)$$

The III-stage IDFT can be easily implemented by using sum & differences of last two stage o/p.



From the above diagram we observe that at each stage we require  $\frac{N}{2}$  multiplication &  $N$  additions. As there are  $\log_2 N$  stages total no. of multiplications is  $\frac{N}{2} \log_2 N$  & total no. of additions is  $N \log_2 N$ .

(P) Compute the IDFT of a sequence  
 $X(k) = \{ 3.5, 0.5, -0.5, 0.5, -0.5, 0.5, -0.5, 0.5 \}$   
 using DIT FFT algorithm.

Soln:-

$$W_N^0 = 1$$

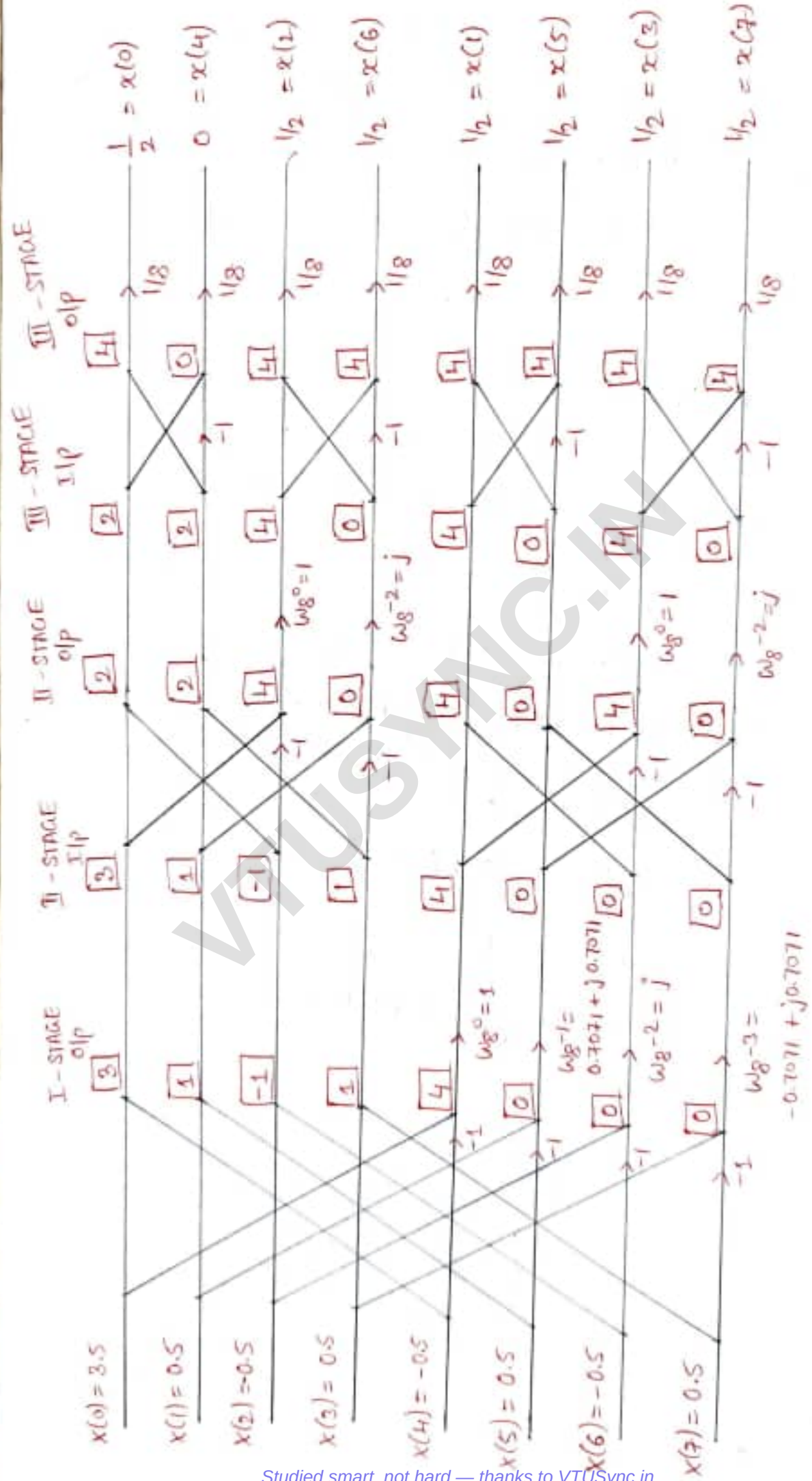
$$W_N^{-1} = 0.7071 + j 0.7071$$

$$W_N^{-2} = j$$

$$W_N^{-3} = -0.7071 + j 0.7071$$

(P.T.O)





$$x(n) = \left\{ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right\}$$

## DIF - FFT ALGORITHM TO COMPUTE IDFT OF

### A SEQUENCE

The IDFT of a  $N$ -point sequence is given by

$$x(m) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-km} \quad \text{--- (1)}$$

Dividing above summation into odd and even parts.

$$x(m) = \frac{1}{N} \left[ \sum_{k=0}^{N/2-1} X(k) (W_N)^{-km} + \sum_{k=1}^{N/2-1} X(k) (W_N)^{-km} \right] \quad \text{--- (2)}$$

Put  $k=2r$  in first summation &  $k=2r+1$  in second summation.

$$\begin{aligned} x(m) &= \frac{1}{N} \left[ \sum_{r=0}^{N/2-1} X(2r) W_N^{-2rm} + \sum_{r=0}^{N/2-1} X(2r+1) W_N^{-(2r+1)m} \right] \\ &= \frac{1}{N} \left[ \sum_{r=0}^{N/2-1} X(2r) W_N^{-2rm} + \sum_{r=0}^{N/2-1} X(2r+1) W_N^{-2rm} W_N^{-m} \right] \end{aligned}$$

changing  $r$  to  $k$ .

$$x(m) = \frac{1}{N} \left[ \sum_{k=0}^{N/2-1} \left[ X(2k) + X(2k+1) W_N^{-m} \right] W_N^{-2km} \right]$$

$$\text{Let } g(m) = \sum_{k=0}^{N/2-1} X(2k) W_N^{-2km}, \quad h(m) = \sum_{k=0}^{N/2-1} X(2k+1) W_N^{-2km} \quad \text{--- (3) and (4)}$$

$$\therefore x(m) = \frac{1}{N} [g(m) + W_N^{-m} h(m)]$$

$$x(0) = \frac{1}{N} [g(0) + W_N^0 h(0)]$$

$$h(m+N) = h(m)$$

$$g(m+N) = g(m)$$

$$x(1) = \frac{1}{N} [g(1) + W_N^{-1} h(1)]$$

$$\therefore x(4) = \frac{1}{N} [g(0) + W_N^{-4} h(0)]$$

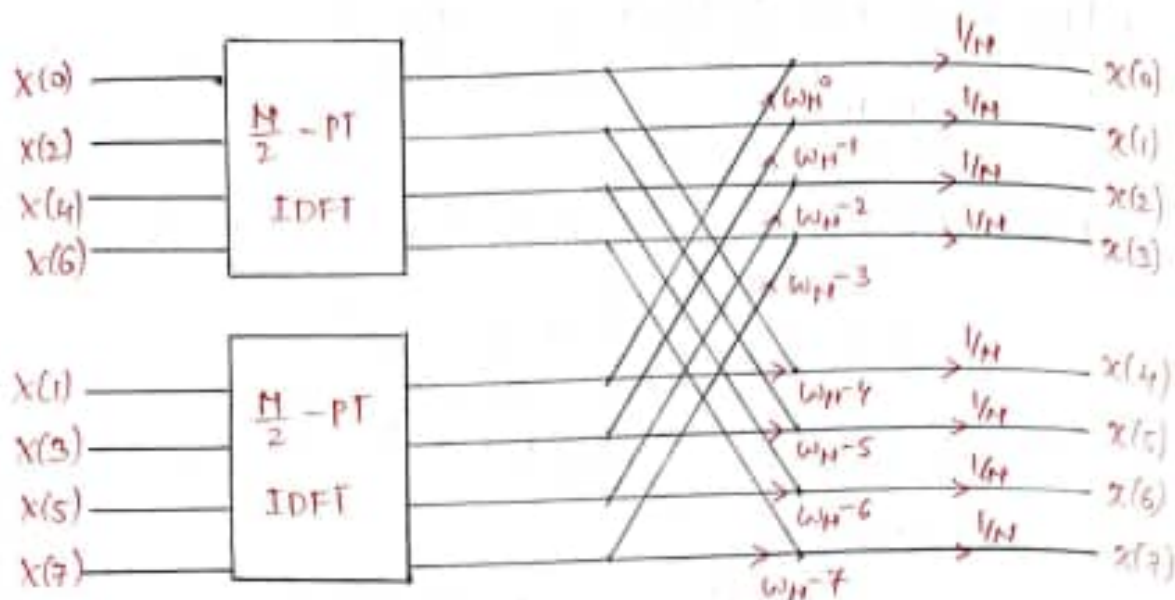
$$x(2) = \frac{1}{N} [g(2) + W_N^{-2} h(2)]$$

$$x(5) = \frac{1}{N} [g(1) + W_N^{-5} h(1)]$$

$$x(6) = \frac{1}{N} [g(2) + W_N^{-6} h(2)]$$

$$x(3) = \frac{1}{N} [g(3) + W_N^{-3} h(3)]$$

$$x(7) = \frac{1}{N} [g(3) + W_N^{-7} h(3)]$$



From ③, 
$$g(m) = \sum_{k=0}^{N/2-1} G(k) (W_{N/2})^{-km}$$

Obtained by replacing  $x(2k)$  by  $G(k)$

Splitting the above equation into odd and even

$$g(m) = \sum_{k=0}^{N/2-2} G(k) (W_{N/2})^{-km} + \sum_{k=1}^{N/2-1} G(k) (W_{N/2})^{-km}$$

taking  $k=2l$  in 1<sup>st</sup> summation &  $k=2l+1$  in 2<sup>nd</sup> summation.

$$g(m) = \sum_{l=0}^{N/4-1} G(2l) (W_{N/2})^{-2lm} + \sum_{l=0}^{N/4-1} G(2l+1) (W_{N/2})^{-(2l+1)m}$$

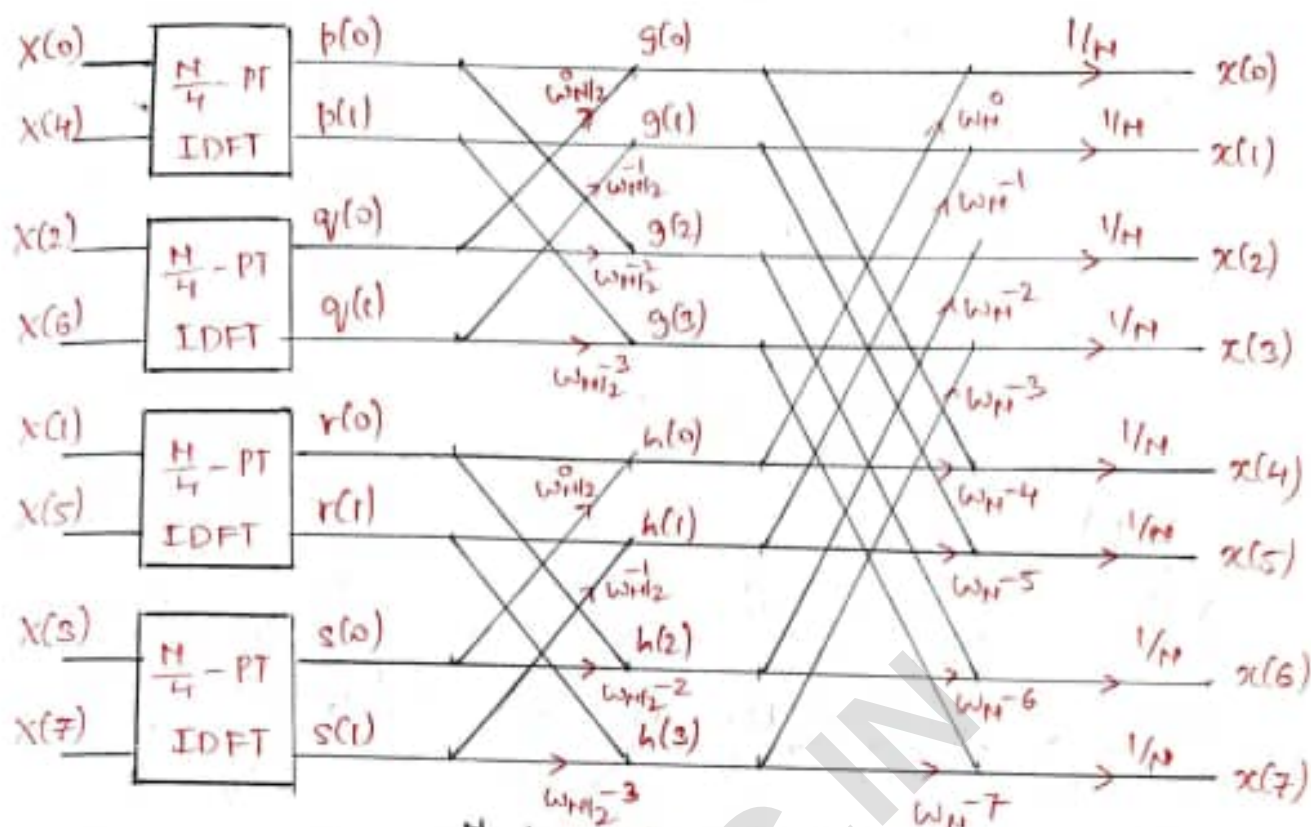
$$= \sum_{l=0}^{N/4-1} G(2l) (W_{N/4})^{-lm} + \sum_{l=0}^{N/4-1} G(2l+1) (W_{N/4})^{-lm} (W_{N/2})^{-l}$$

Let 
$$p(m) = \sum_{l=0}^{N/4-1} G(2l) W_{N/4}^{-lm}$$

$$q(m) = \sum_{l=0}^{N/4-1} G(2l+1) (W_{N/4})^{-lm}$$

$$\therefore g(m) = p(m) + W_{N/2}^{-m} q(m)$$

IIIy 
$$h(m) = r(m) + W_{N/2}^{-m} s(m)$$

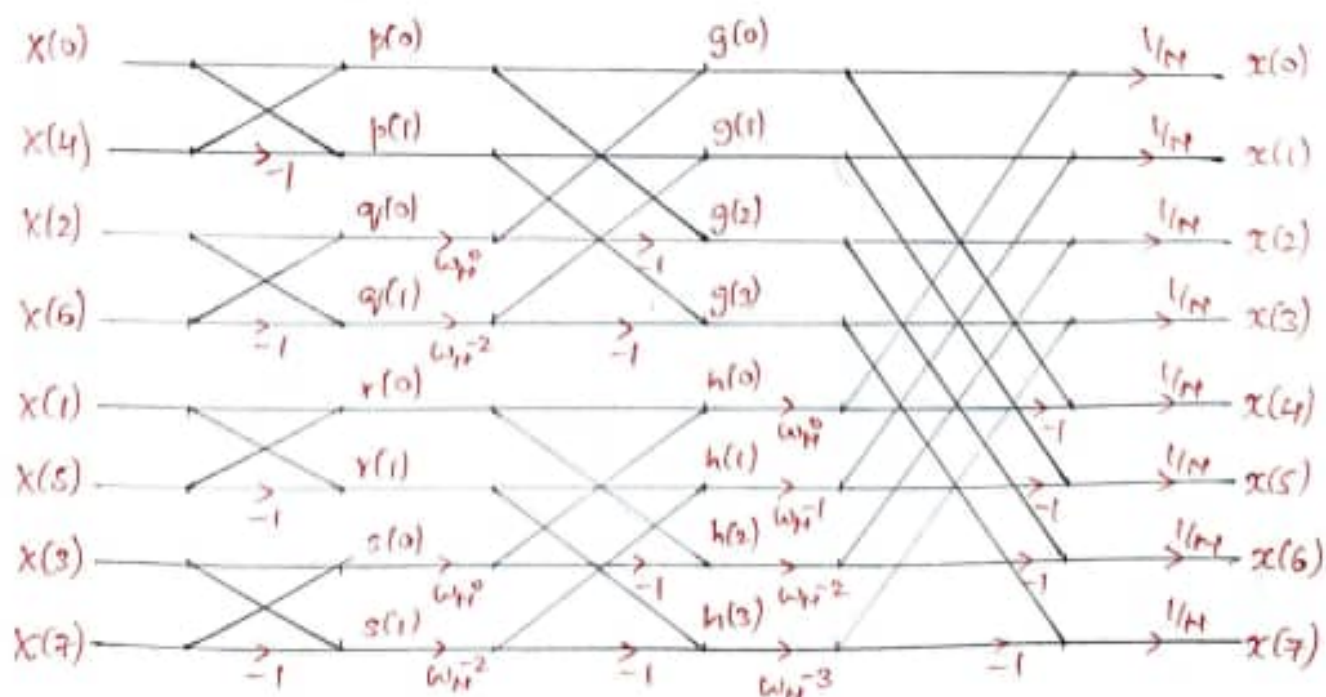


$$p(m) = \sum_{k=0}^{\frac{N}{4}-1} X(4k) W_N^{-km}$$

$$p(0) = X(0) W_2^0 + X(4) W_2^0 = X(0) + X(4)$$

$$p(1) = X(0) W_2^0 + X(4) W_2^{-1} = X(0) - X(4)$$

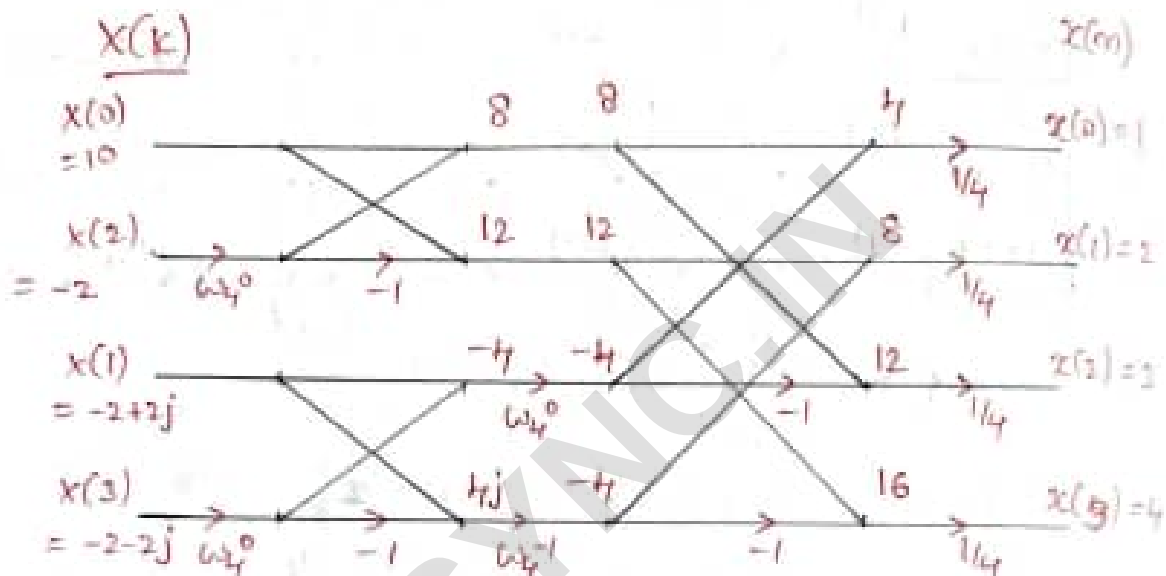
$$W_N^{-(m+N/2)} = -W_N^{-m}$$





① Compute IDFT of  $X(k) = \{10, -2+2j, -2, -2-2j\}$  using DIF-FFT algorithm.

Soln: -  $\omega_4^0 = 1$   
 $\omega_4^{-1} = j$



$\therefore x(m) = \underline{\underline{\{1, 2, 3, 4\}}}$



## DFT used as IDFT

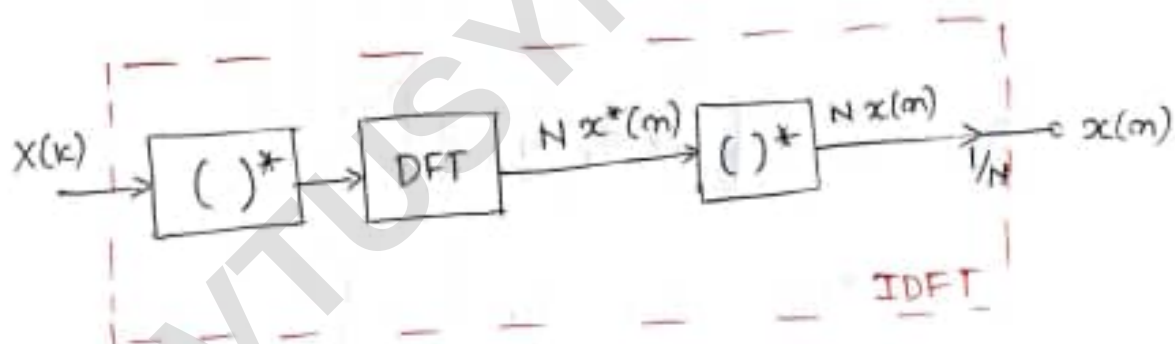
$$\text{DFT : } X(k) = \sum_{m=0}^{N-1} x(m) W_N^{km}$$

$$\text{IDFT : } x(m) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-km}$$

Taking complex conjugate of IDFT eqn,

$$x^*(m) = \frac{1}{N} \sum_{k=0}^{N-1} X^*(k) W_N^{km}$$

$$x^*(m) = \frac{1}{N} \text{DFT} \{ X^*(k) \}$$



- ① Take complex conjugate of  $X(k)$ .
- ② Compute DFT using DFT equation.
- ③ Take complex conjugate of result
- ④ Divide by  $N$ .

① Compute IDFT of  $X(k) = \{4, 2j, 0, -2j\}$  using DFT.

Soln:-  $X(k) = \{4, 2j, 0, -2j\}$

$$X^*(k) = \{4, -2j, 0, 2j\}$$

$$\text{DFT} \{X^*(k)\} = Nx^*(m)$$

$$= \begin{bmatrix} 4 & -2j & 0 & 2j \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$Nx^*(m) = [4, 0, 4, 8]$$

$$\therefore x^*(m) = [1, 0, 1, 2]$$

$$x(m) = \underline{\underline{[1, 0, 1, 2]}}$$

② Compute IDFT of  $X(k) = \{10, -2+2j, -2, -2-2j\}$  using DFT.

Soln:-

$$x^*(k) = \{10, -2-2j, -2, -2+2j\}$$

$$\text{DFT} \{x^*(k)\} = N x^*(m)$$

$$= [10, -2-2j, -2, -2+2j] \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$= [4, 8, 12, 16]$$

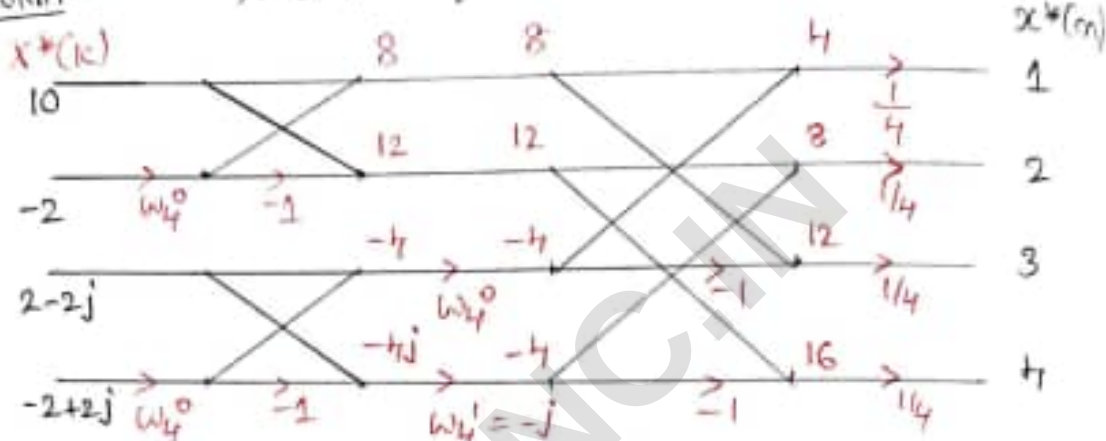
$$x^*(m) = [1, 2, 3, 4]$$

$$\therefore x(m) = [1, 2, 3, 4]$$

# Computing IDFT of a sequence using DFT - FFT Algorithm

- ① Compute IDFT of  $X(k) = \{10, -2+2j, -2, -2-2j\}$  using DIT-FFT / DFT Algorithm.

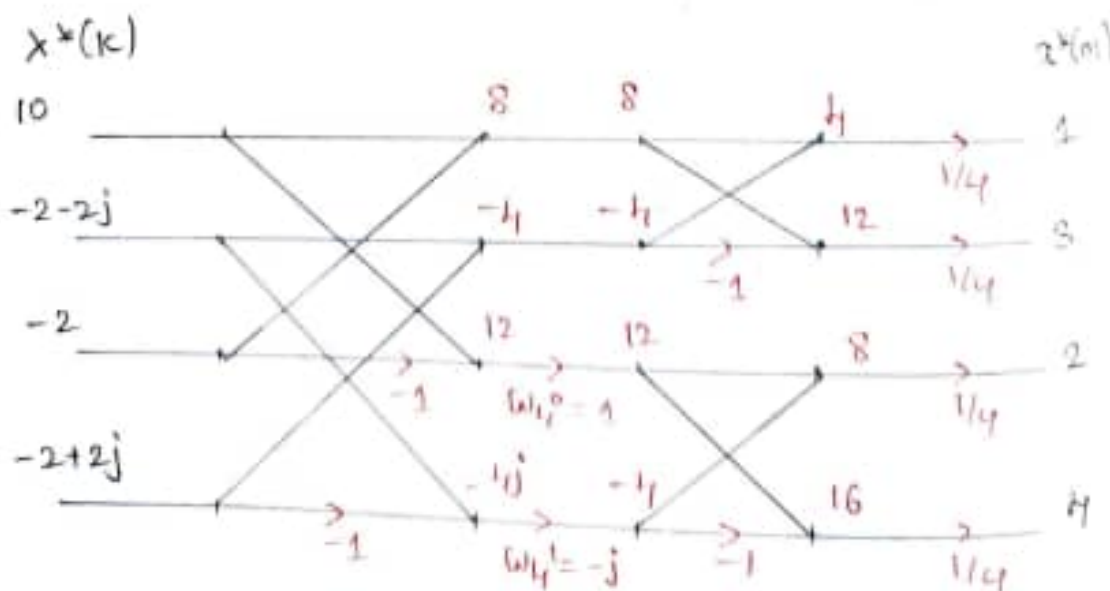
Soln:-



$$\therefore x(m) = \{1, 2, 3, 4\}$$

- ② Compute IDFT of  $X(k) = \{10, -2+2j, -2, -2-2j\}$  using DIF-FFT / DFT algorithm.

$$X^*(k) = \{10, -2-2j, -2, -2+2j\}$$



# Comparison of Computational Complexity for the Direct Computation of the DFT versus the FFT Algorithm

| Number of Points, $N$ | Complex multiplications in Direct Computation $N^2$ | Complex multiplications in FFT Algorithms $(N/2) \log_2 N$ | Speed Improvement Factor $\frac{N^2}{(N/2) \log_2 N}$ |
|-----------------------|-----------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------|
| 4                     | 16                                                  | 4                                                          | 4.0                                                   |
| 8                     | 64                                                  | 12                                                         | 5.3                                                   |
| 16                    | 256                                                 | 32                                                         | 8.0                                                   |
| 32                    | 1,024                                               | 80                                                         | 12.8                                                  |
| 64                    | 4,096                                               | 192                                                        | 21.3                                                  |
| 128                   | 16,384                                              | 448                                                        | 36.6                                                  |
| 256                   | 65,536                                              | 1,024                                                      | 64.0                                                  |
| 512                   | 262,144                                             | 2,304                                                      | 113.8                                                 |
| 1,024                 | 1,048,576                                           | 5,120                                                      | 204.8                                                 |

## Inplace computation:-

Once a butterfly operation is performed on a pair of complex numbers  $(a, b)$  to produce  $(A, B)$ , there is no need to save the input pair  $(a, b)$ . Hence we store the result  $(A, B)$  in the same location as  $(a, b)$ . Consequently, we require a fixed amount of storage, namely,  $2N$  storage registers in order to store



the results ( $N$  complex numbers) of the computations at each stage.

Since the same  $2N$  storage locations are used throughout the computation of the  $N$ -point DFT, we say that the computations are done in place.

- ⑨ What is in-place computation? What is the total number of complex additions and multiplications required for  $N=512$  points, if DFT is computed directly and if FFT is used? Also find the number of stages required and its memory requirement. (5) DEC'12

|          | COMPLEX<br>MULTIPLICATIONS     | COMPLEX<br>ADDITIONS |
|----------|--------------------------------|----------------------|
| DIRECT : | $N^2 = 262,144$                | $N(N-1) = 261632$    |
| FFT :    | $\frac{N}{2} \log_2 N = 2,304$ | $N \log_2 N = 4608$  |

$$\begin{aligned} \text{No. of Stages in FFT} &= \log_2 N = \log_2 512 \\ &= 9 \\ &= \end{aligned}$$

$$\begin{aligned} \text{Memory Requirement} &= 2N = 2 \times 512 \\ &= 1024 \end{aligned}$$

$$\begin{aligned} \text{Memory with Twiddle Factor} &= 2N + \frac{N}{2} \\ &= 1024 + 256 \\ &= 1280 \end{aligned}$$